

CSE4203: Computer Graphics

Lecture – 9 (part - A)

# **Ray Tracing**

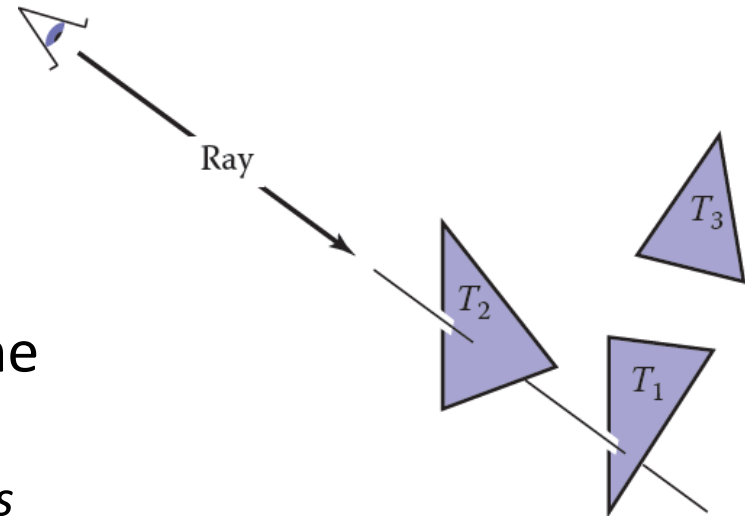
Chapter 4

# Outline

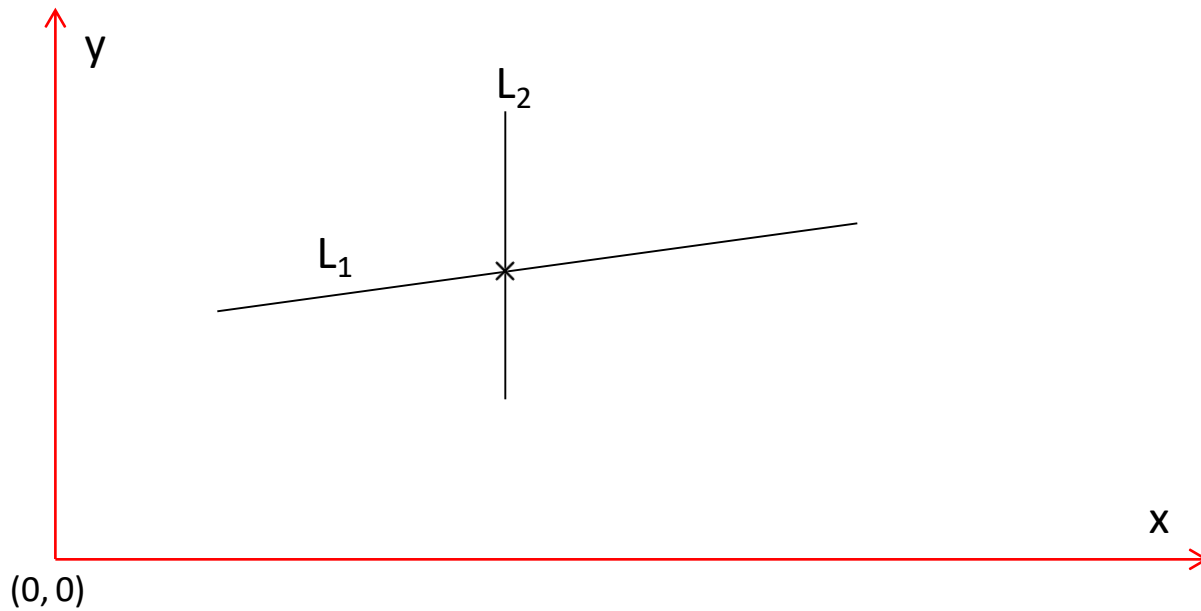
- Ray-tracing

# 3D $\rightarrow$ 2D

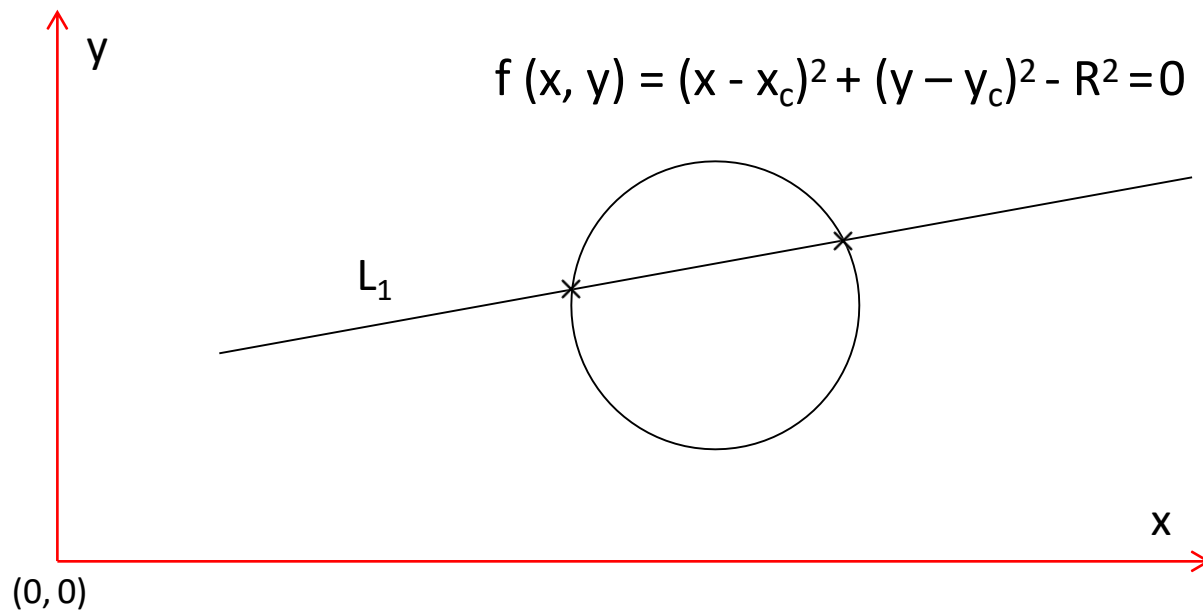
- Implementing projection: (3D  $\rightarrow$  2D)
  - Ray-tracing technique
- Motivation:
  - From how we see!
  - The **ray** is “**traced**” into the **scene**
  - the first **object** hit is the one seen.
    - In this case, the triangle  $T_2$  is returned.



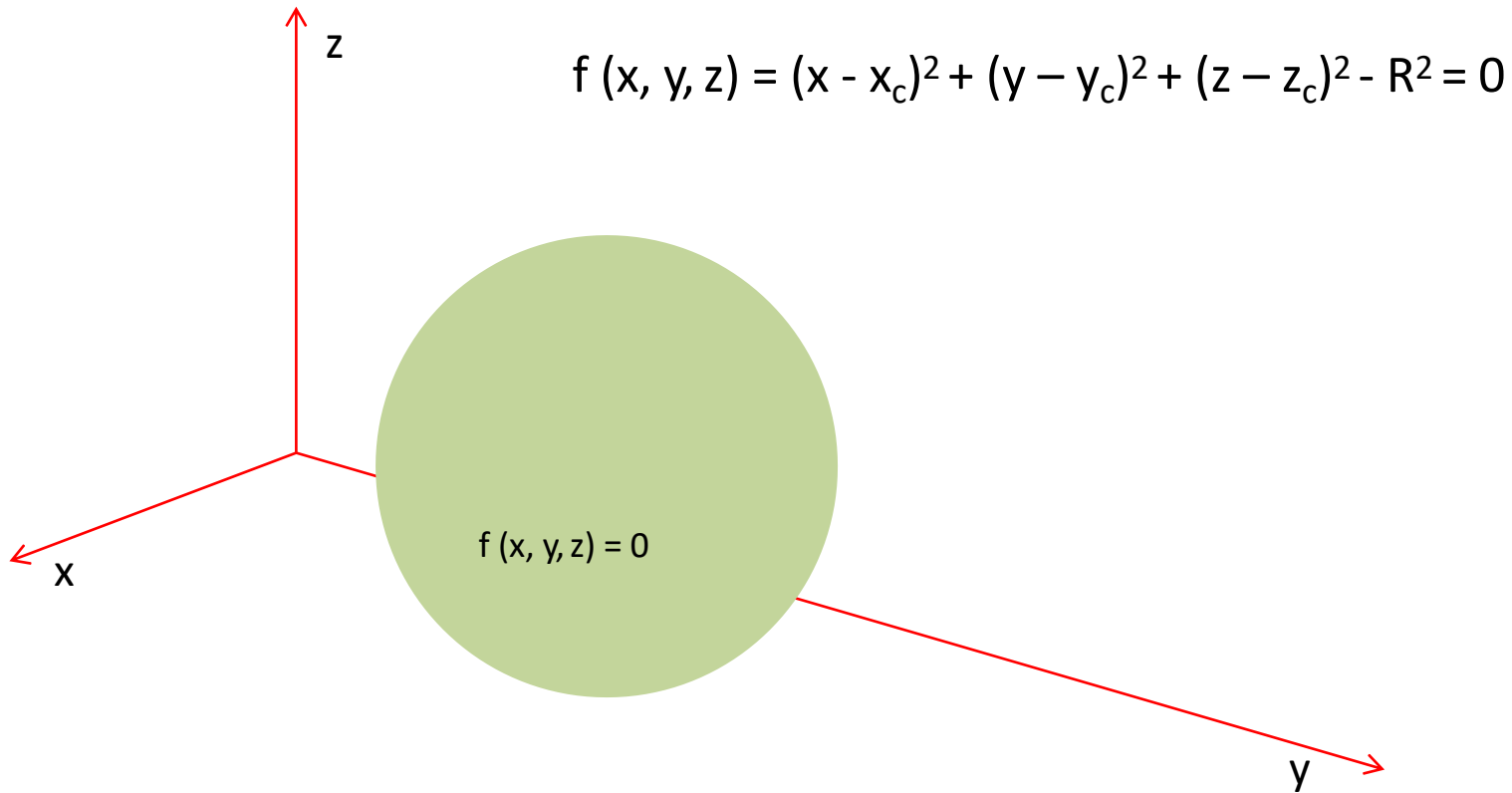
# Warm-up (1/9)



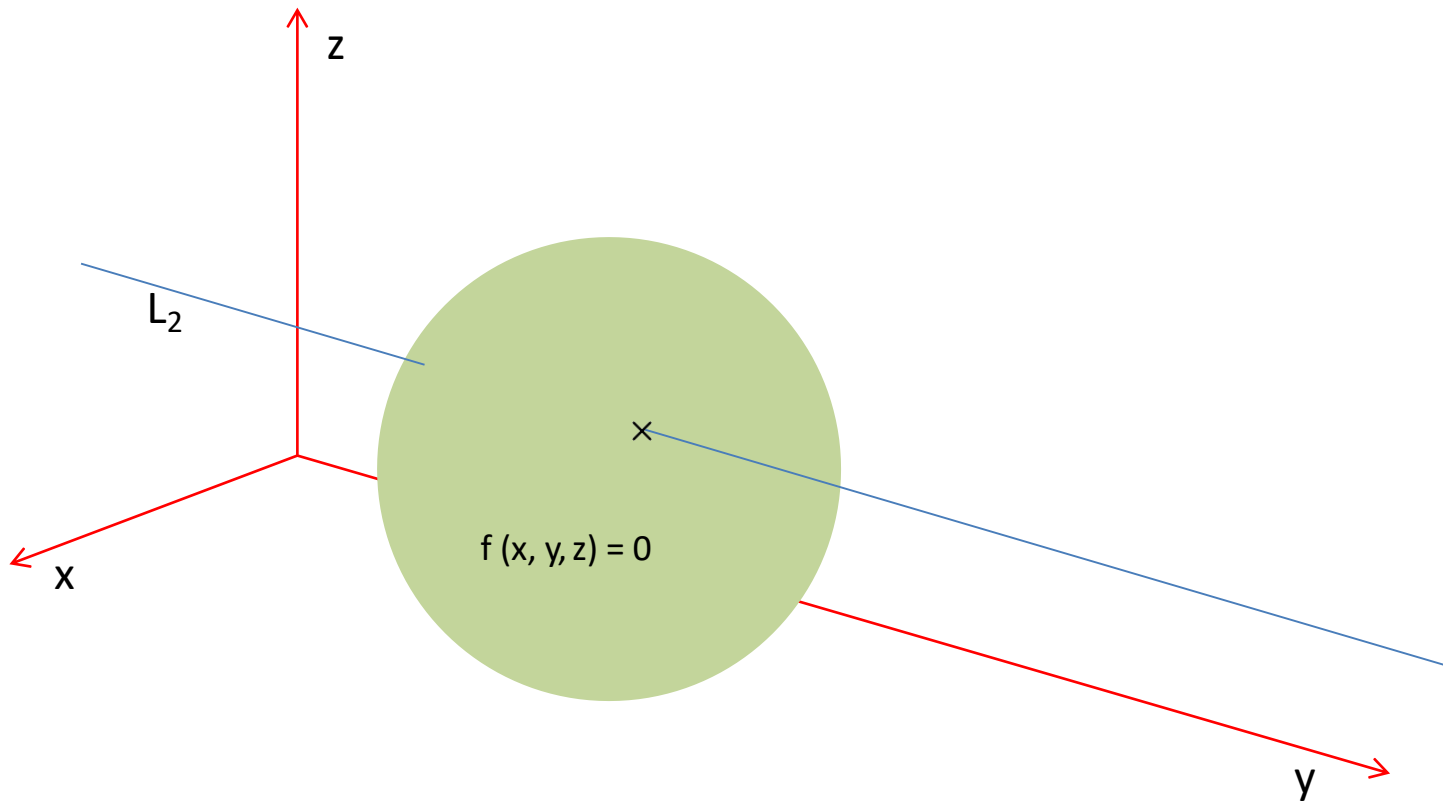
# Warm-up (2/9)



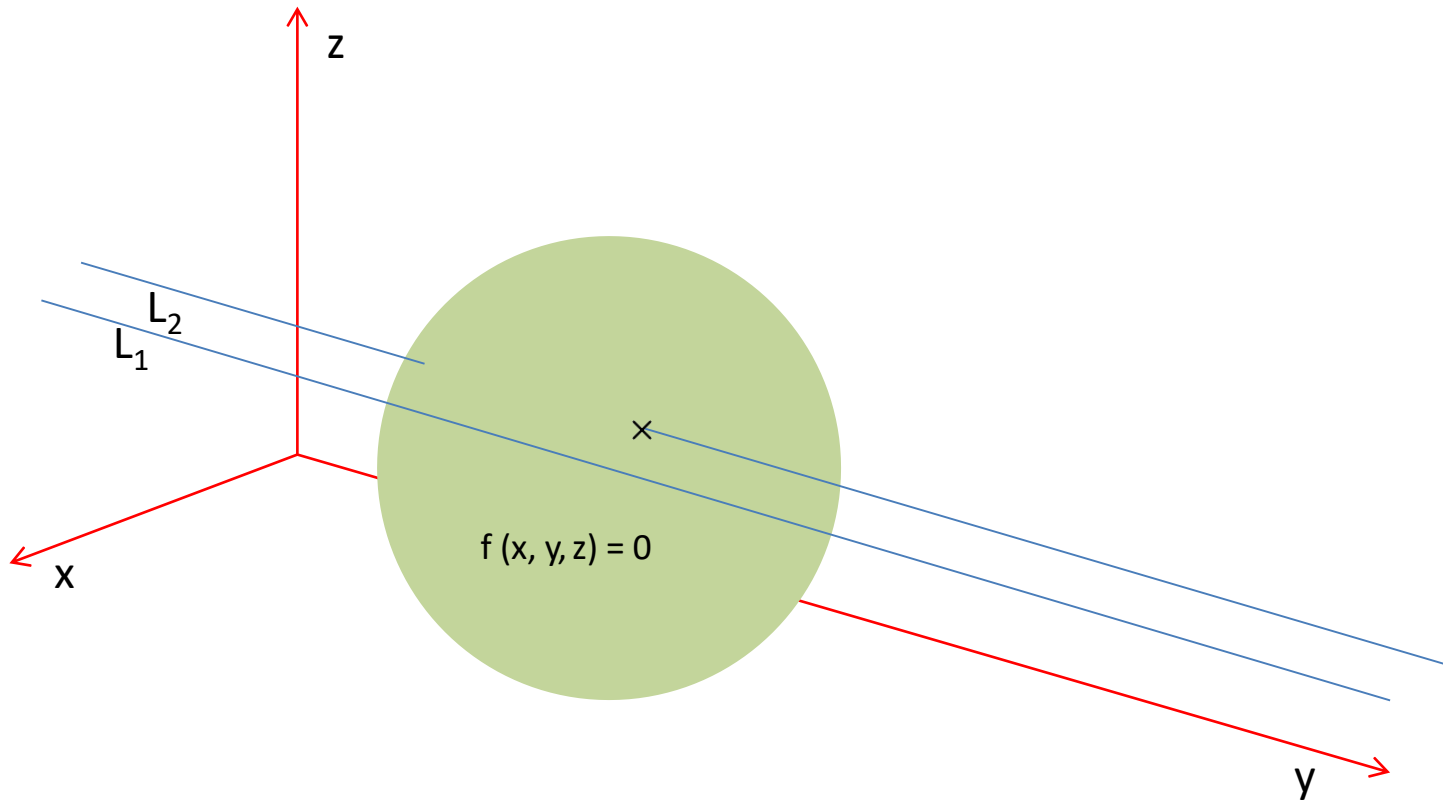
# Warm-up (3/9)



# Warm-up (4/9)

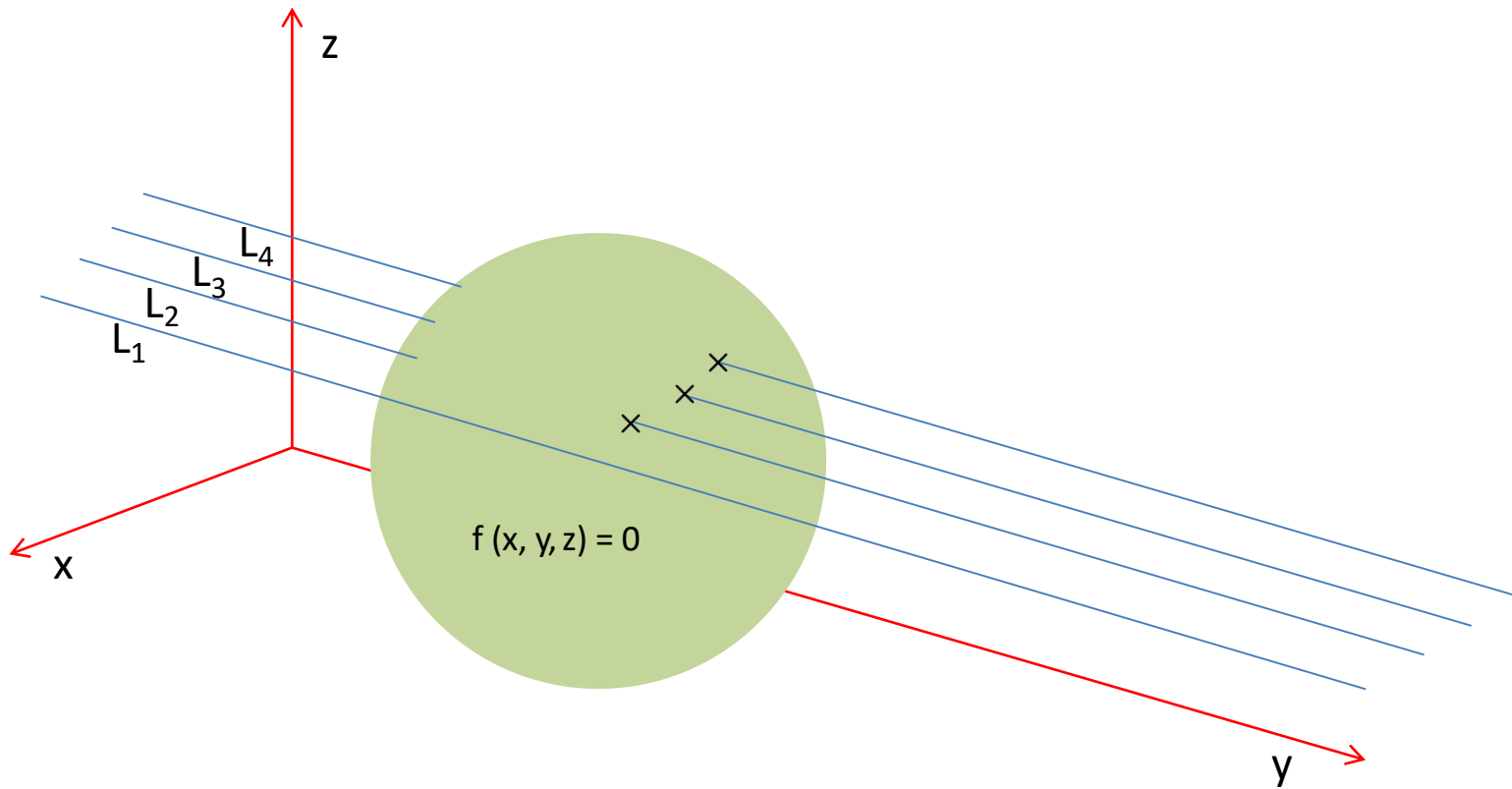


# Warm-up (5/9)

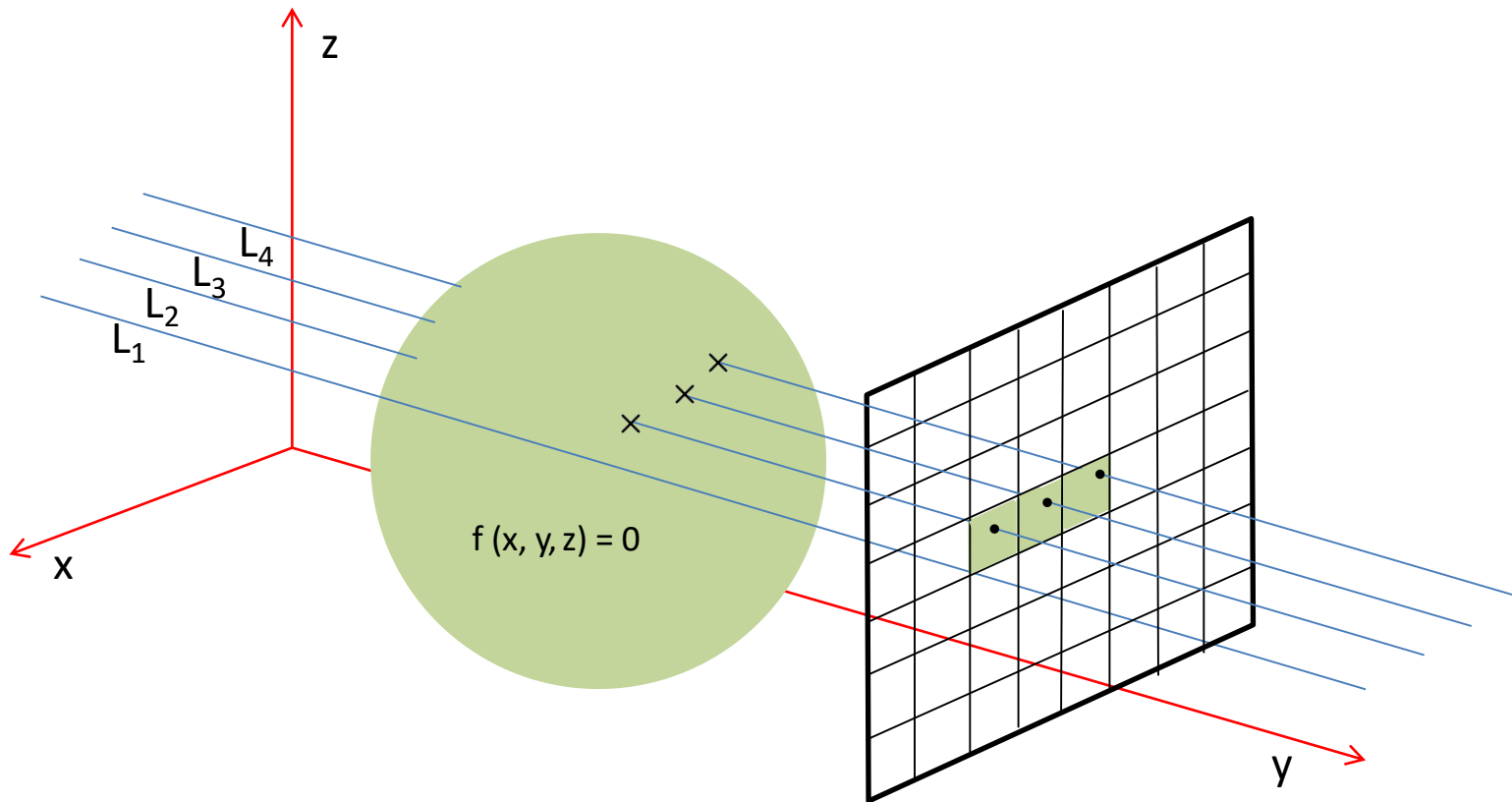




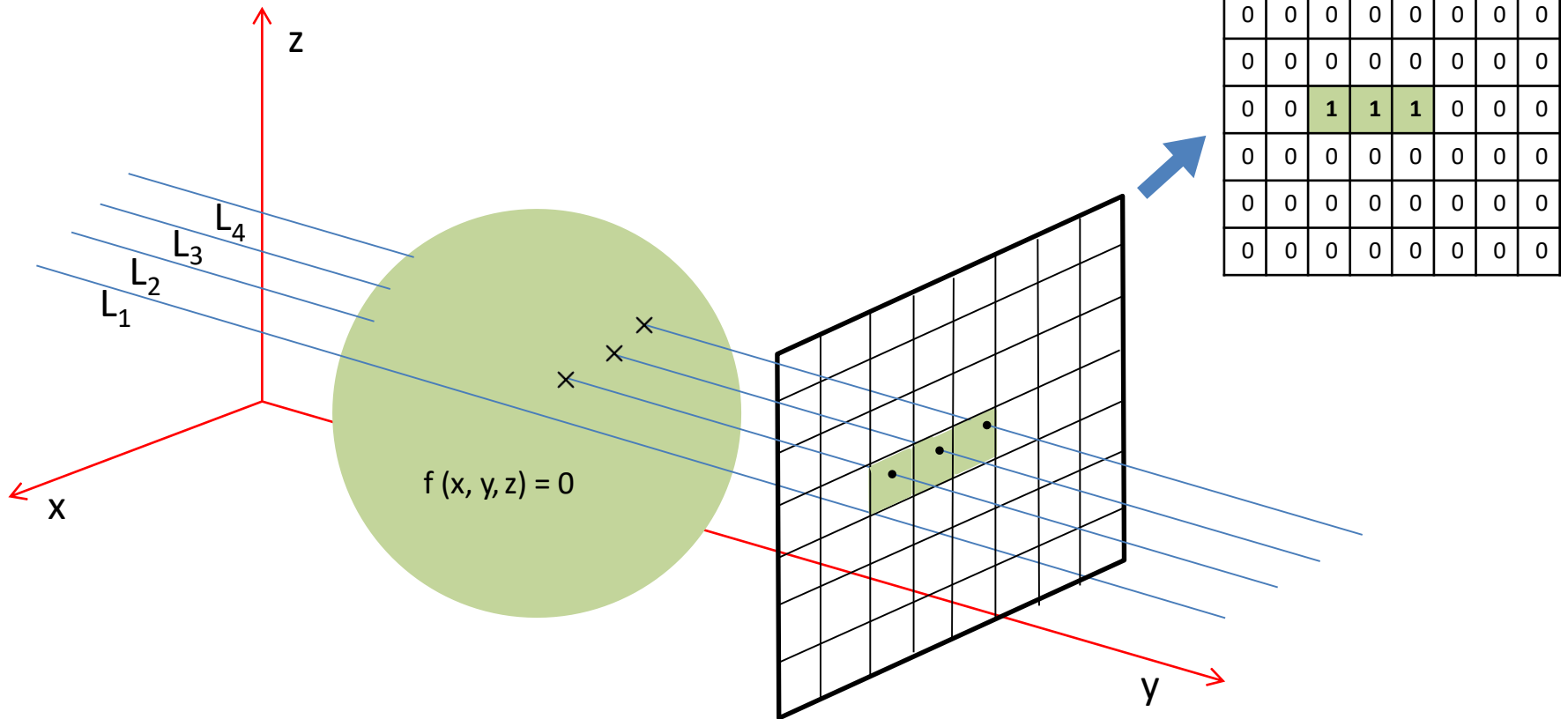
# Warm-up (6/9)



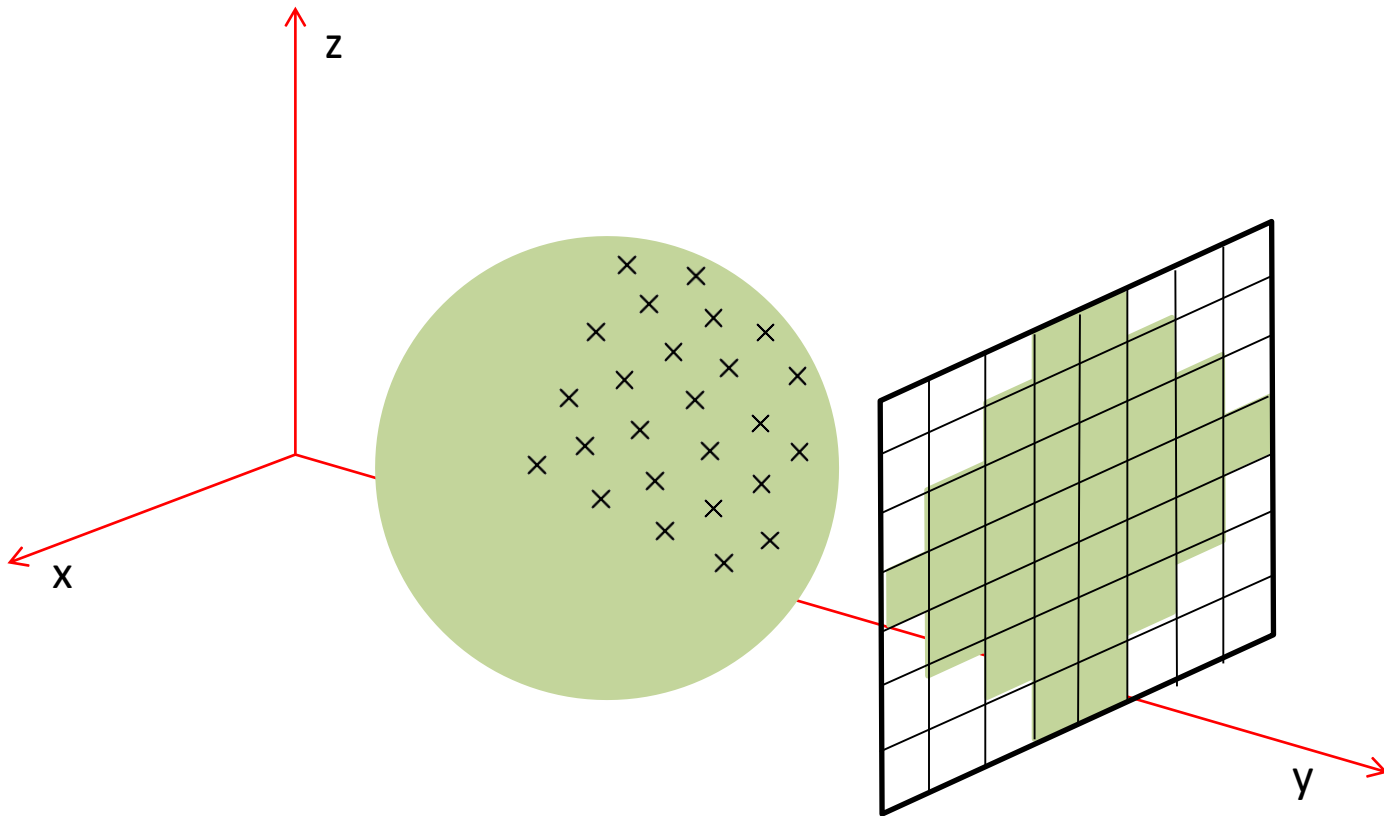
# Warm-up (7/9)



# Warm-up (8/9)



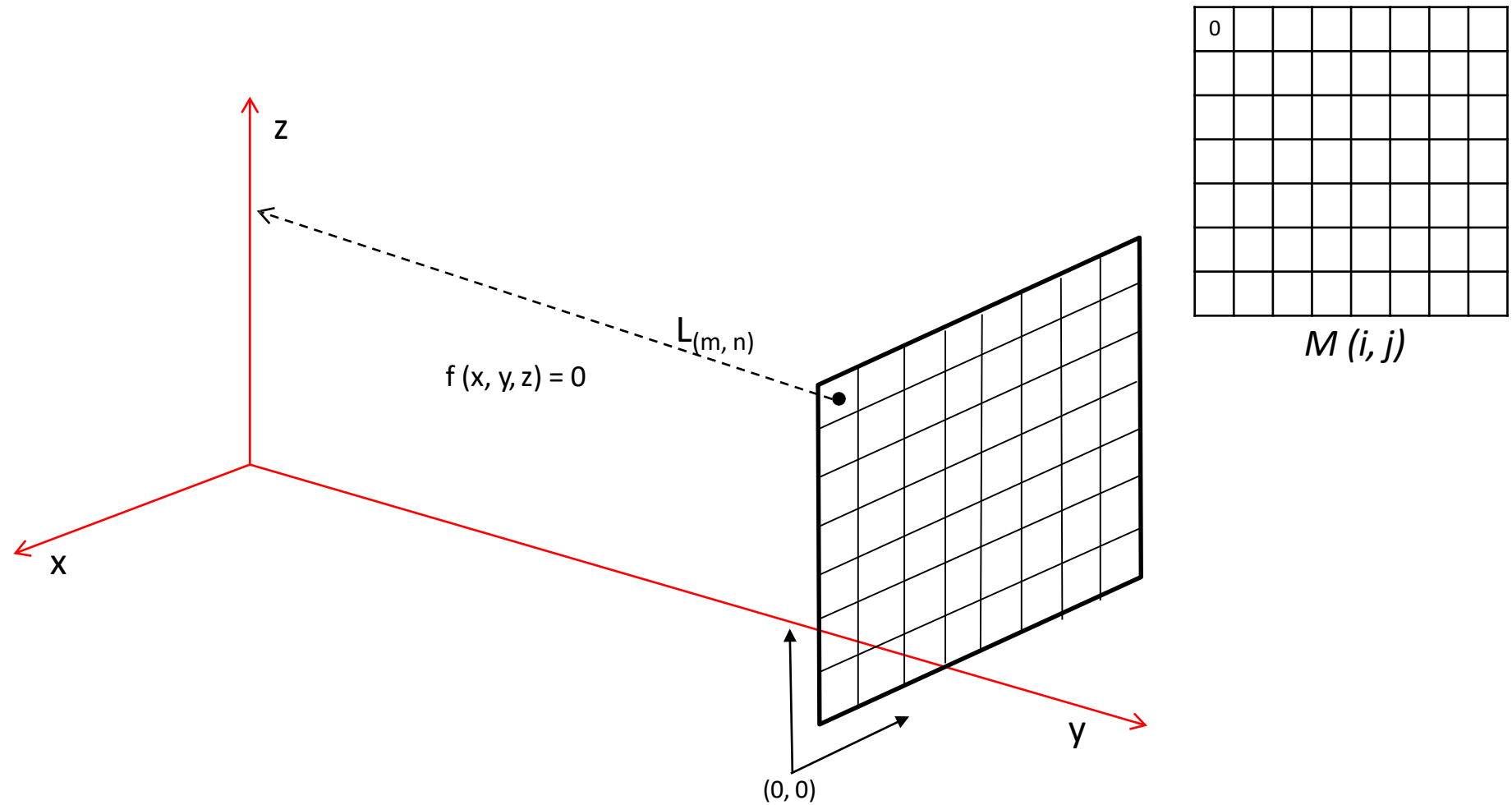
# Warm-up (9/9)



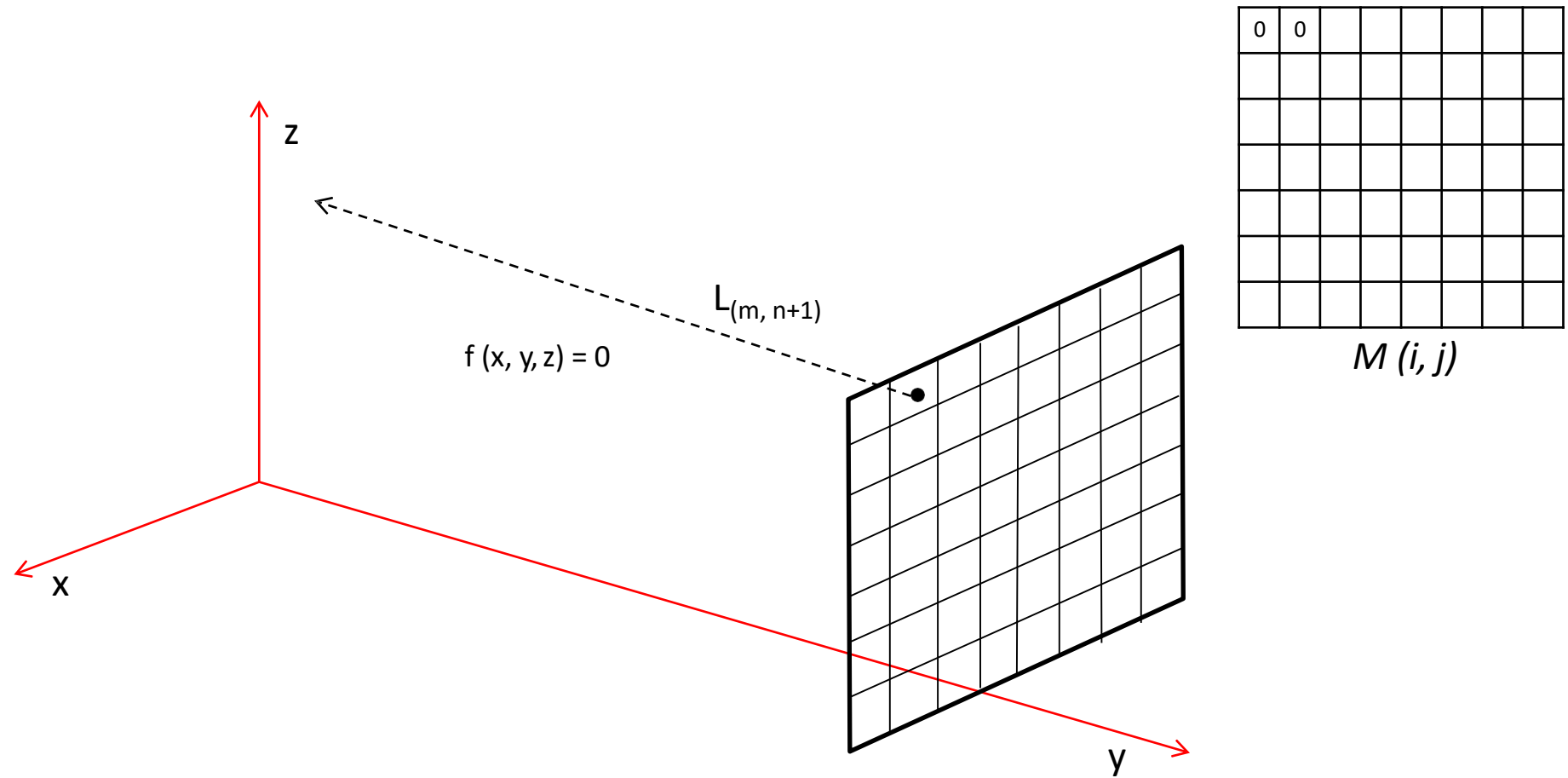
$M(i, j)$

|   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 1 | 1 | 0 | 0 | 0 |
| 0 | 0 | 1 | 1 | 1 | 1 | 0 | 0 |
| 0 | 1 | 1 | 1 | 1 | 1 | 1 | 0 |
| 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| 0 | 1 | 1 | 1 | 1 | 1 | 1 | 0 |
| 0 | 0 | 1 | 1 | 1 | 1 | 0 | 0 |
| 0 | 0 | 0 | 1 | 1 | 0 | 0 | 0 |

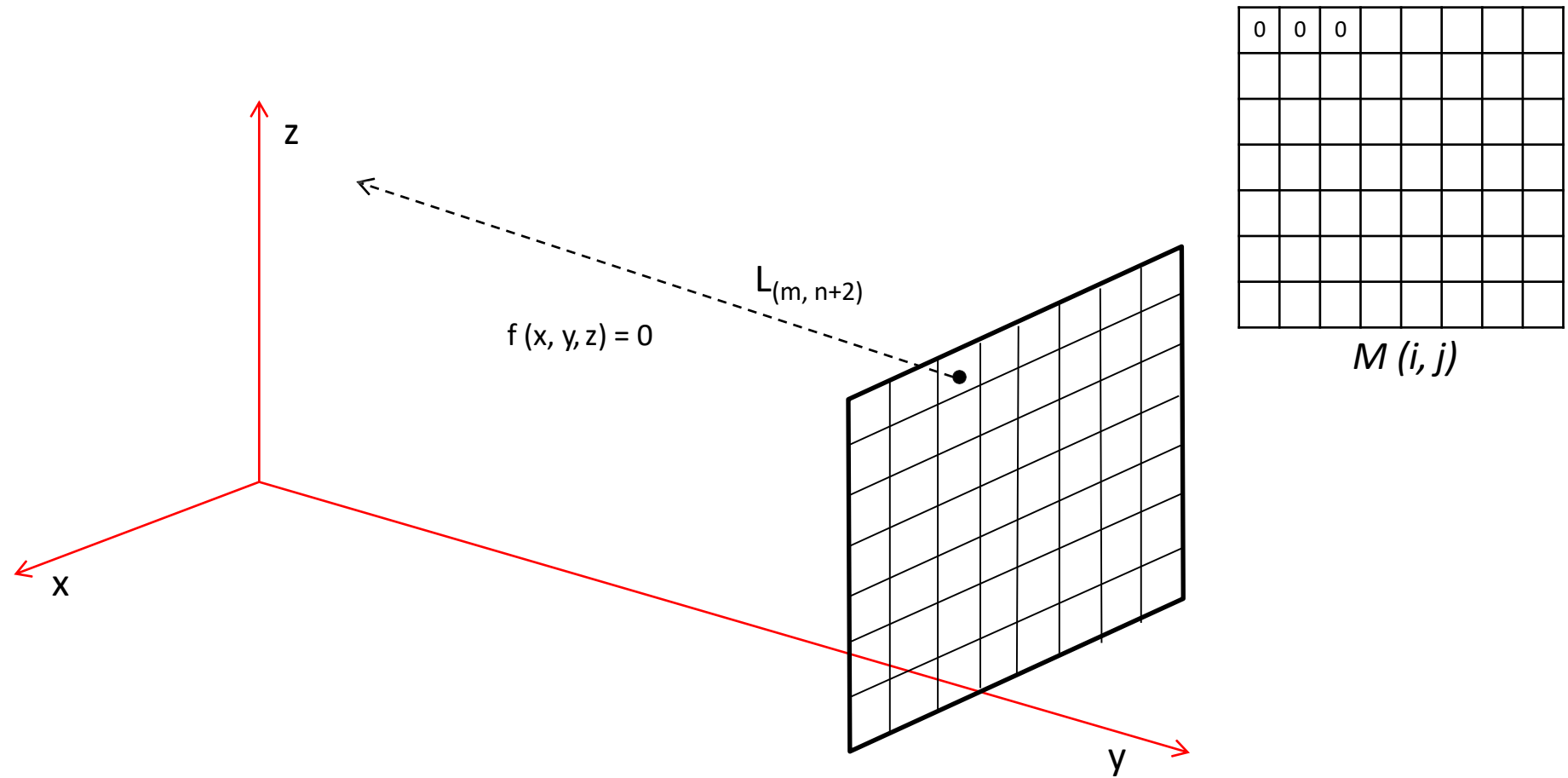
# Ray-tracing Basics (1/15)



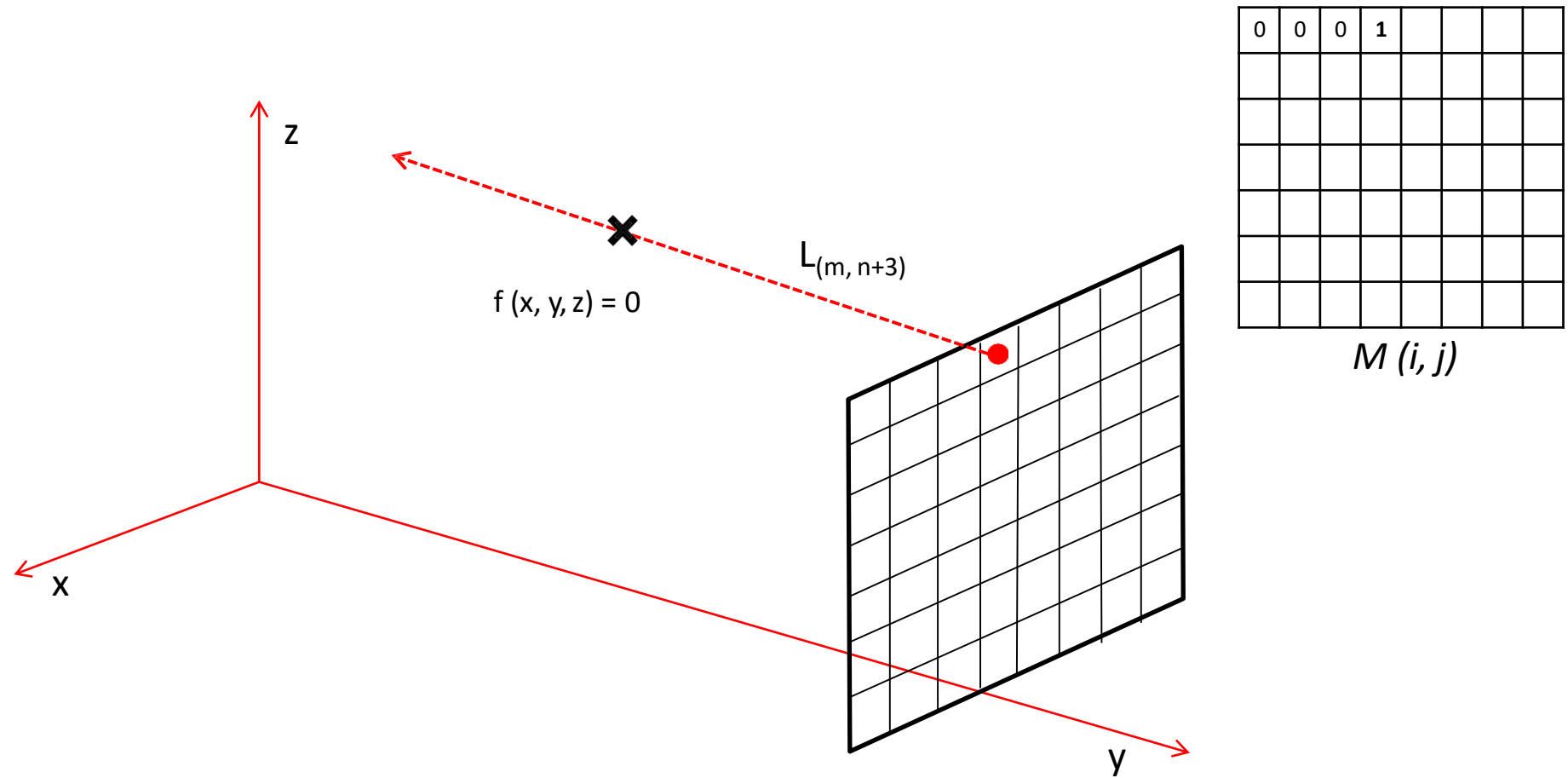
# Ray-tracing Basics (2/15)



# Ray-tracing Basics (3/15)

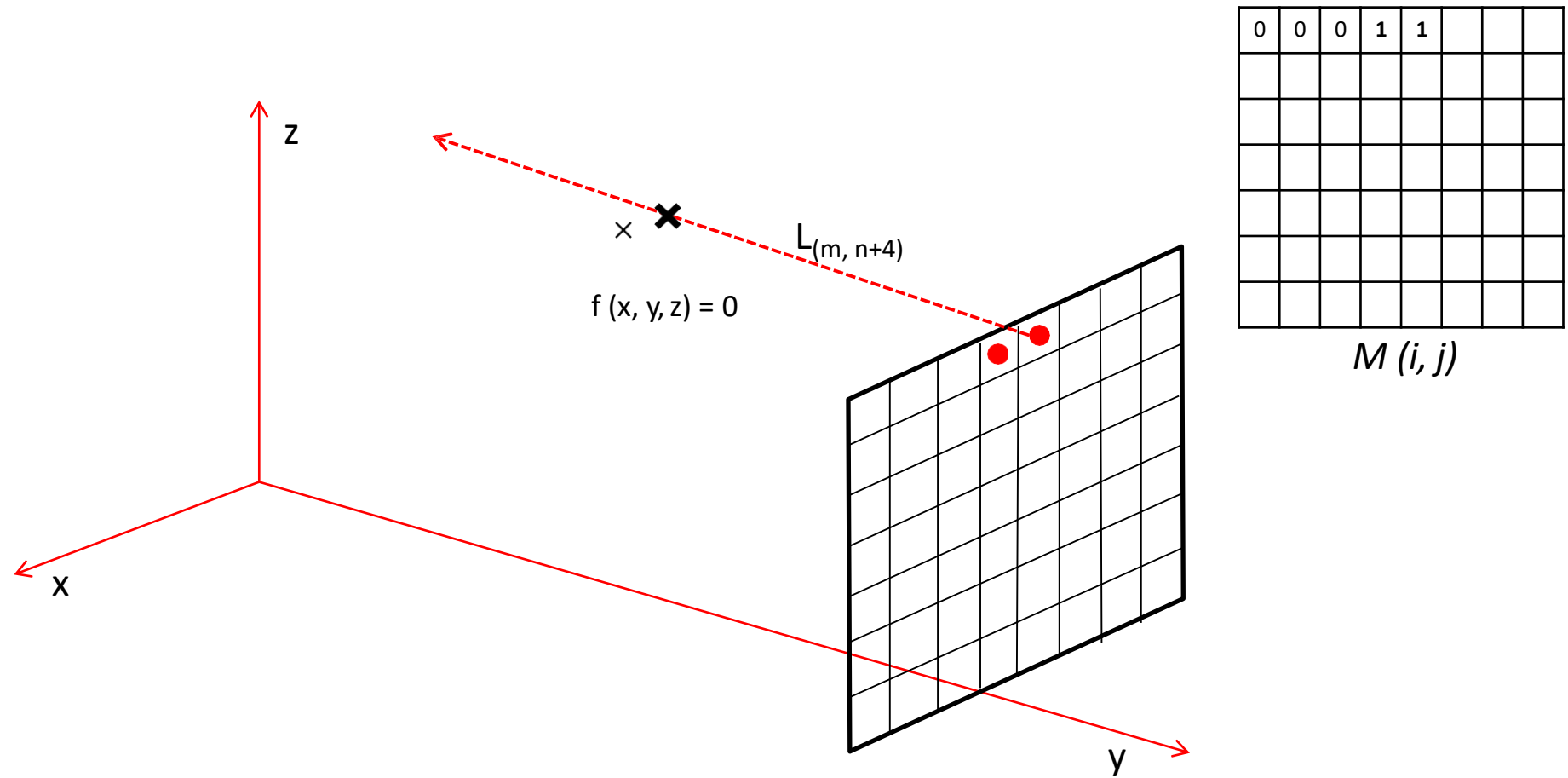


# Ray-tracing Basics (4/15)

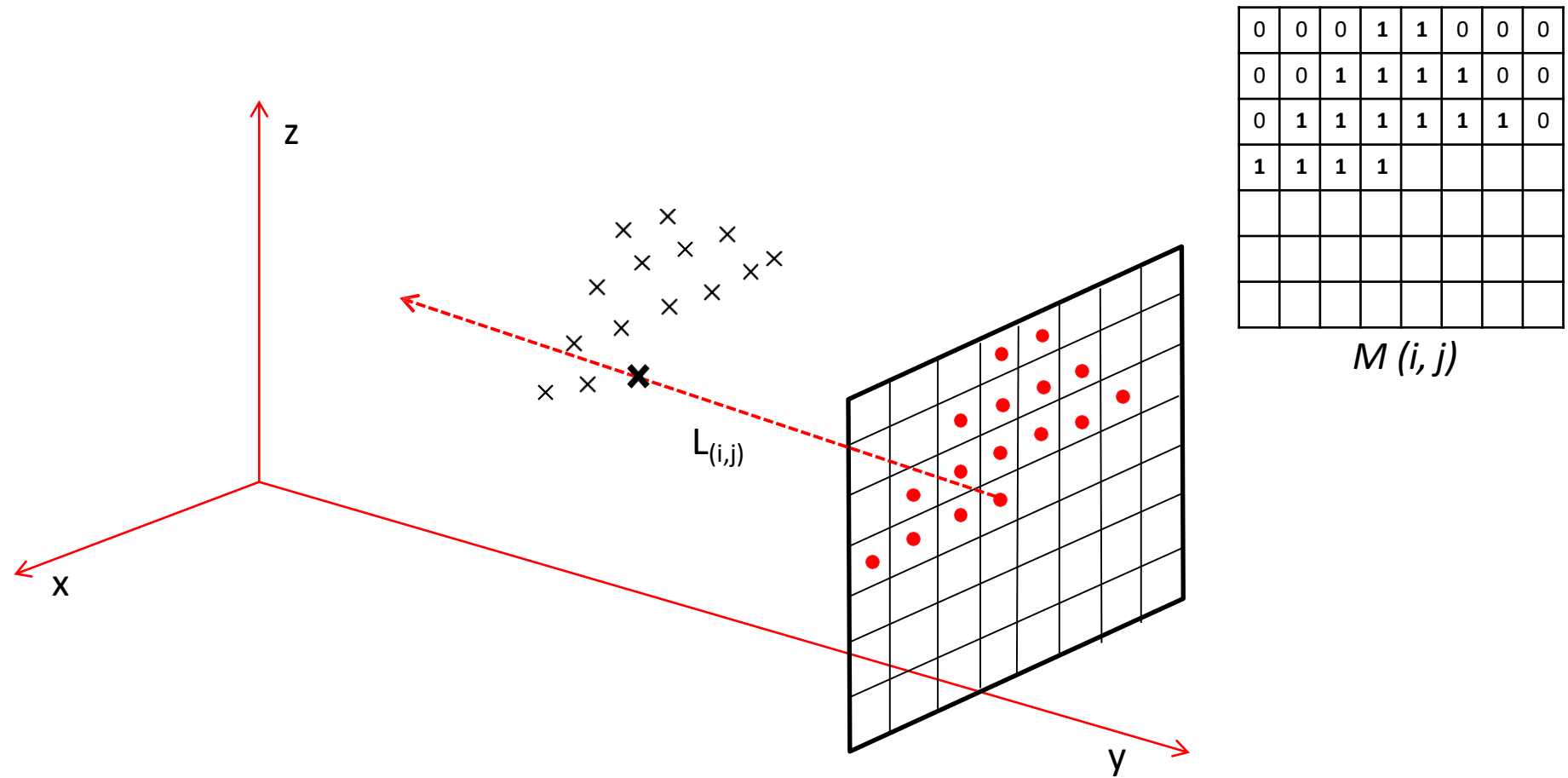




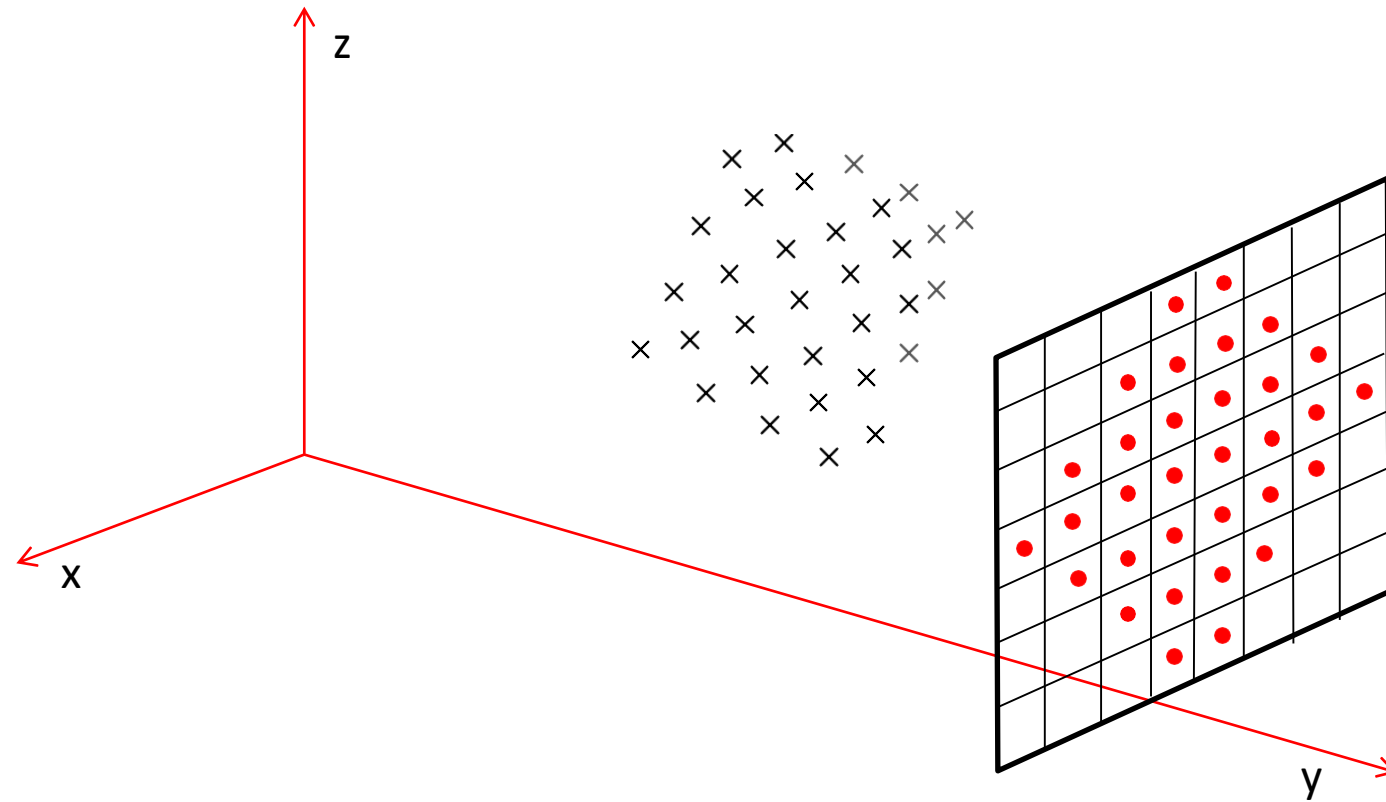
# Ray-tracing Basics (5/15)



# Ray-tracing Basics (6/15)



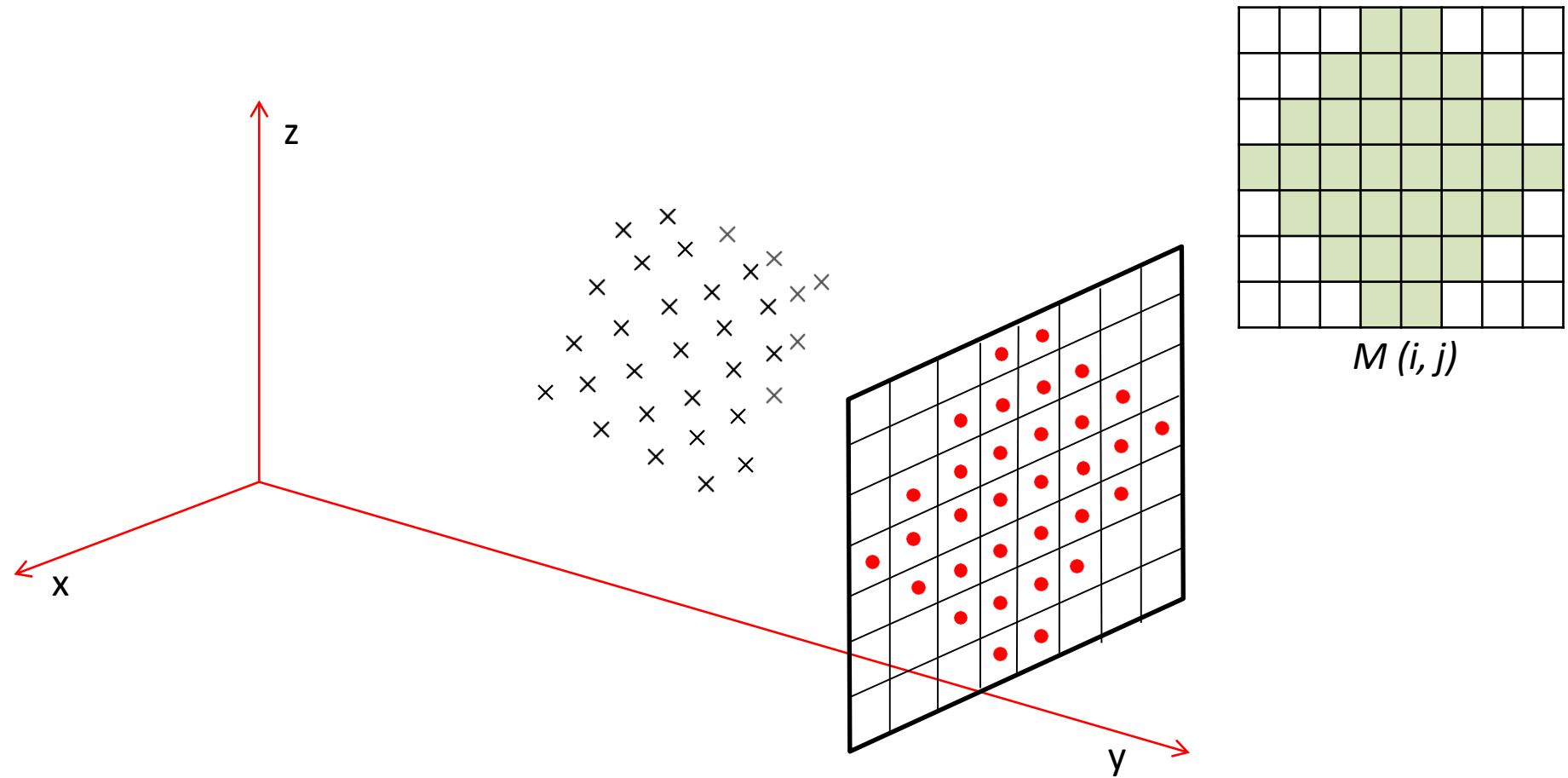
# Ray-tracing Basics (7/15)



|   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 1 | 1 | 0 | 0 | 0 |
| 0 | 0 | 1 | 1 | 1 | 1 | 0 | 0 |
| 0 | 1 | 1 | 1 | 1 | 1 | 1 | 0 |
| 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| 0 | 1 | 1 | 1 | 1 | 1 | 1 | 0 |
| 0 | 0 | 1 | 1 | 1 | 1 | 0 | 0 |
| 0 | 0 | 0 | 1 | 1 | 0 | 0 | 0 |

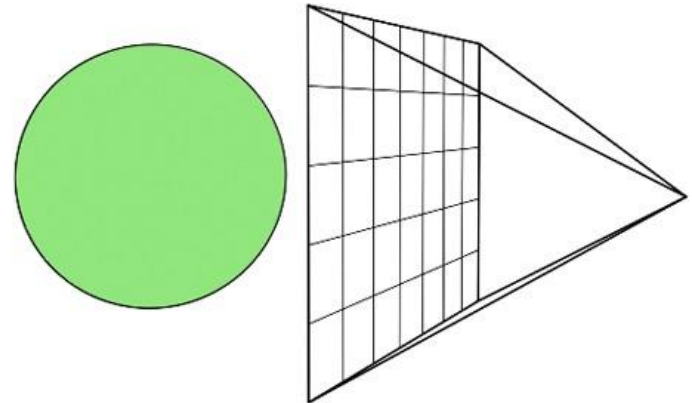
$M(i, j)$

# Ray-tracing Basics (8/15)



# Ray-tracing Basics (9/15)

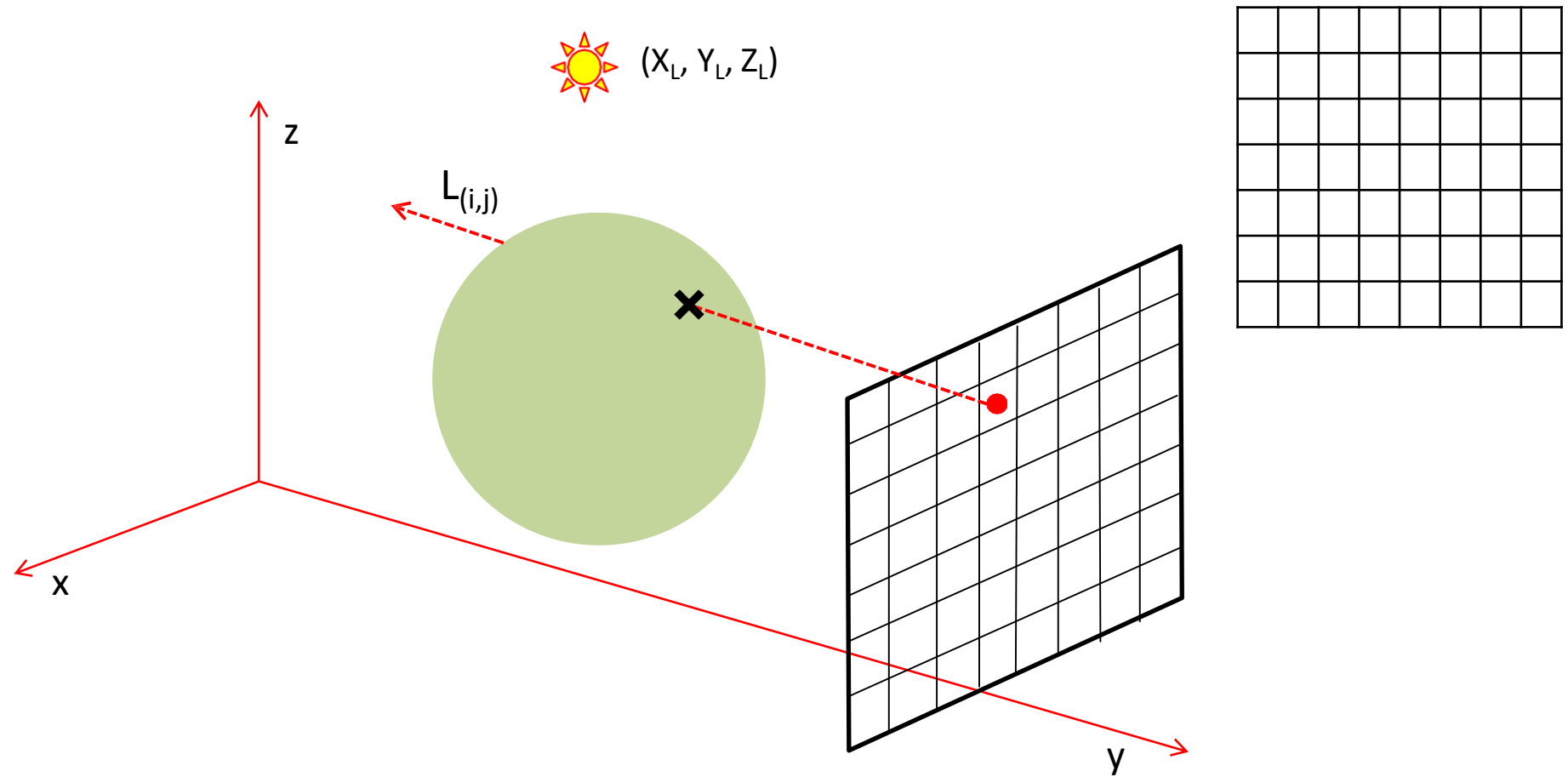
- Computing one pixel at a time
  - Each pixel “**looks**” in a **direction**
- Any object that is seen by a pixel
  - intersect “**viewing ray**”
  - viewing ray: line through that pixel is looking



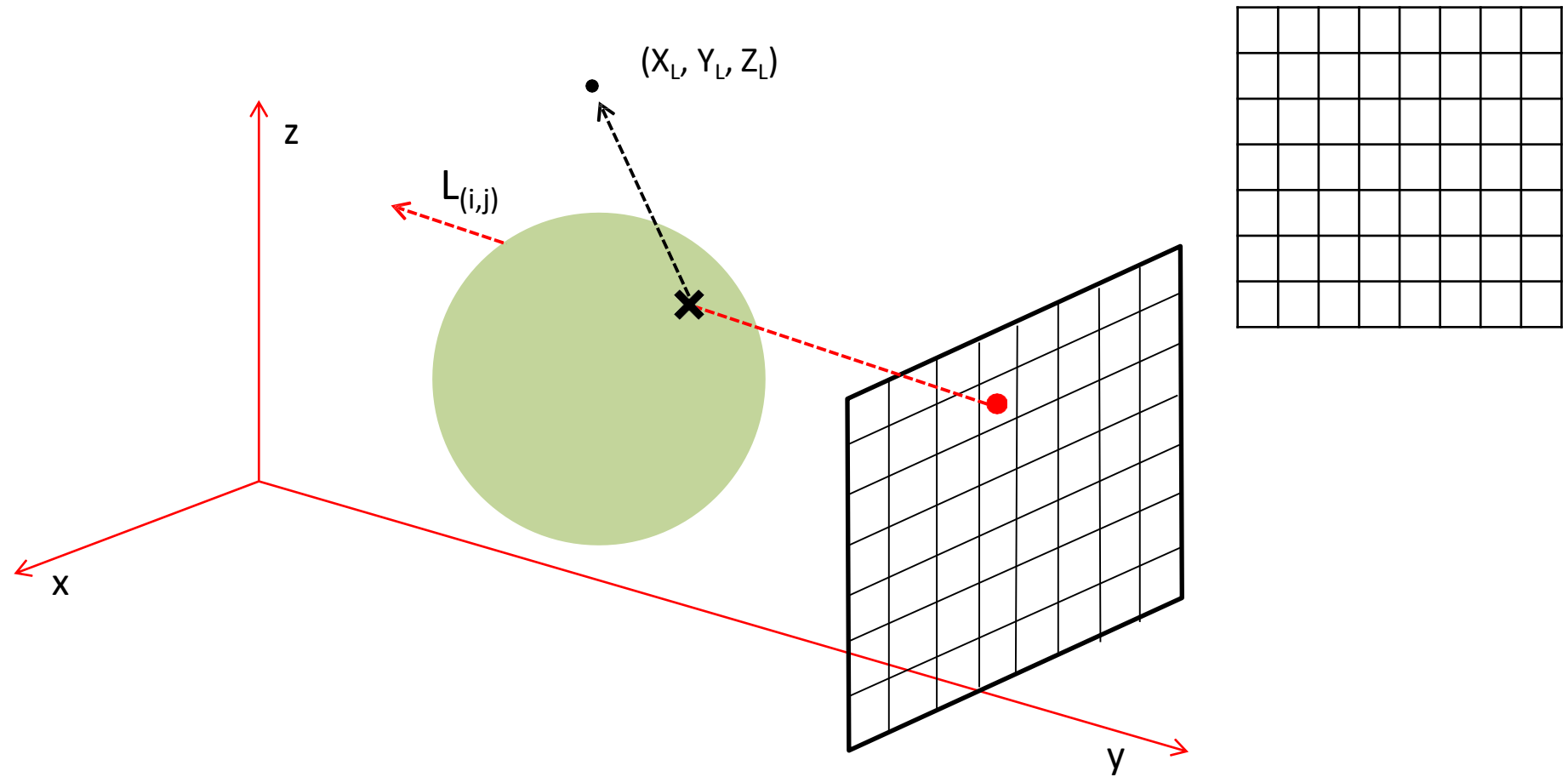
# Ray-tracing Basics (10/15)

- Once that object is found, determine the color of the pixel.
  - a **shading computation** is need, that uses
    - the intersection point
    - surface normal ( $n$ )
    - other information

# Ray-tracing Basics (11/15)

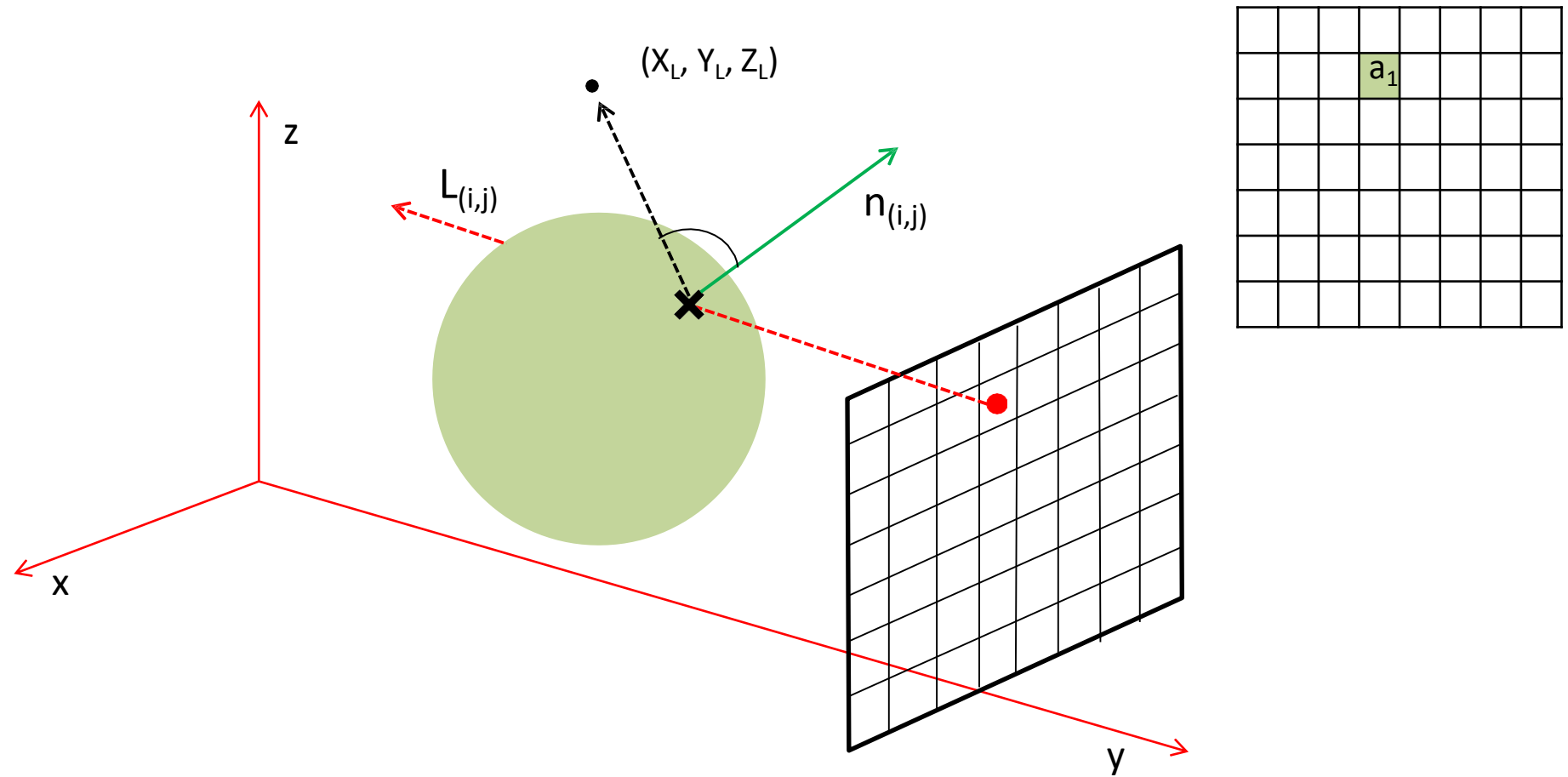


# Ray-tracing Basics (12/15)

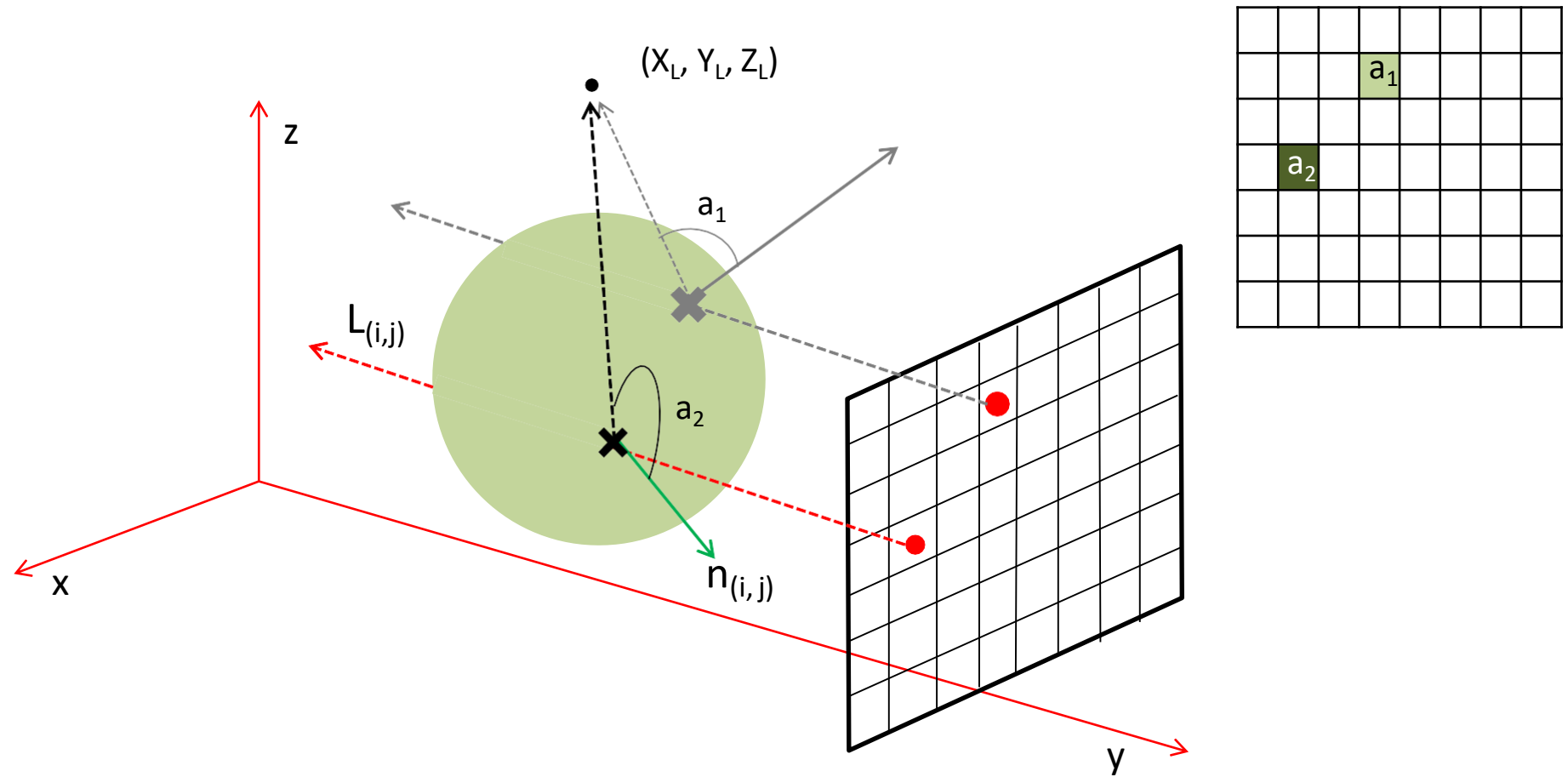




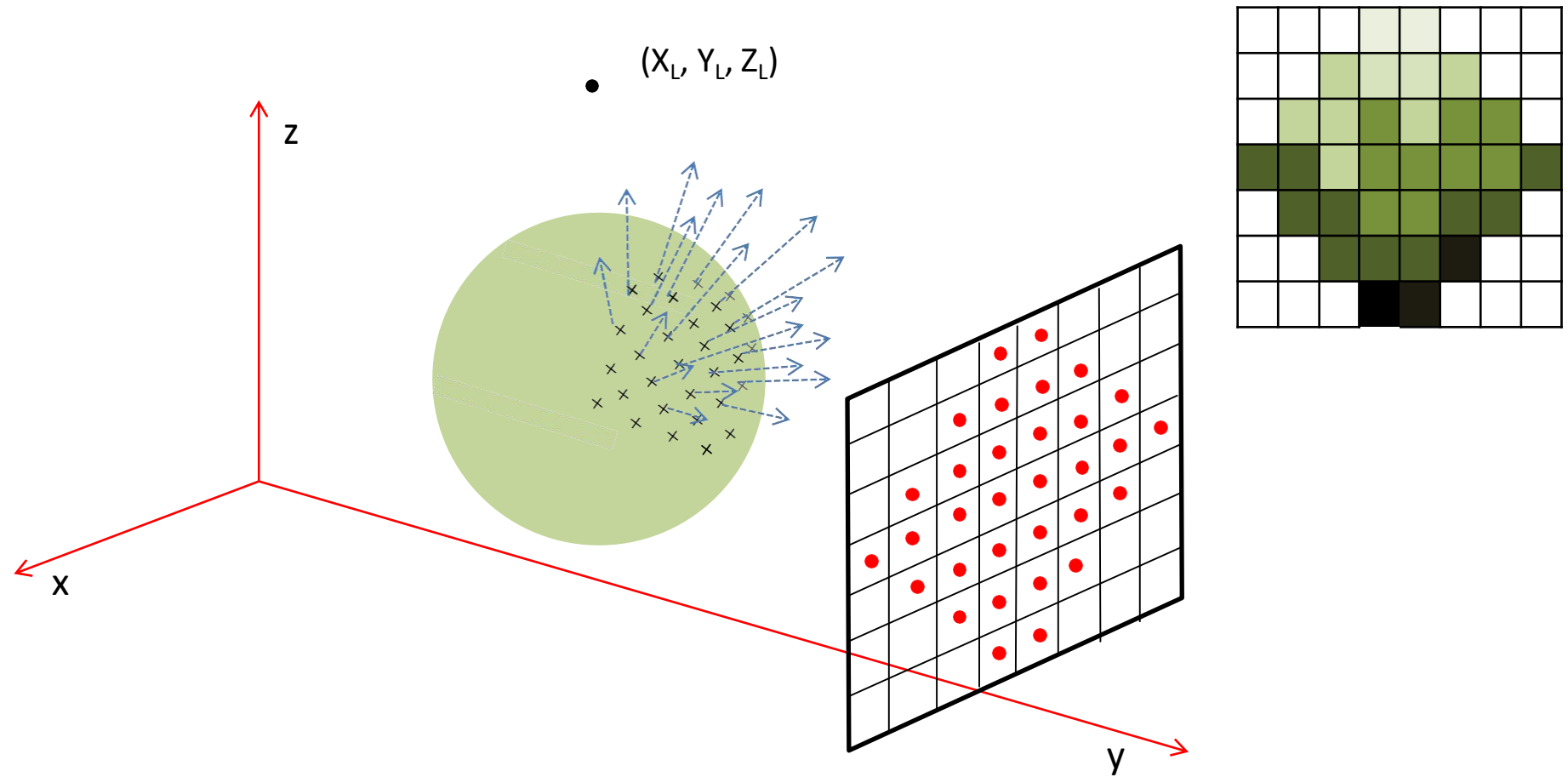
# Ray-tracing Basics (13/15)



# Ray-tracing Basics (14/15)



# Ray-tracing Basics (15/15)

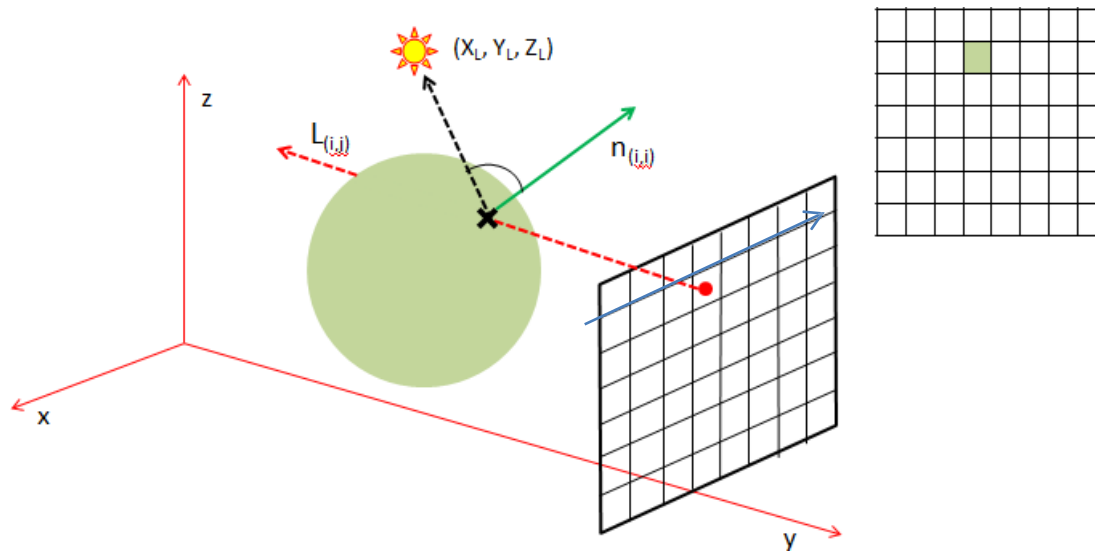


# Ray-Tracing Algorithm (1/3)

- A basic ray tracer therefore has three parts:
  - **ray generation:**
    - computes the origin and direction of each pixel's viewing ray.
  - **ray intersection:**
    - finds the closest object intersecting the viewing ray.
  - **shading:**
    - computes the pixel color based on the results of ray intersection.

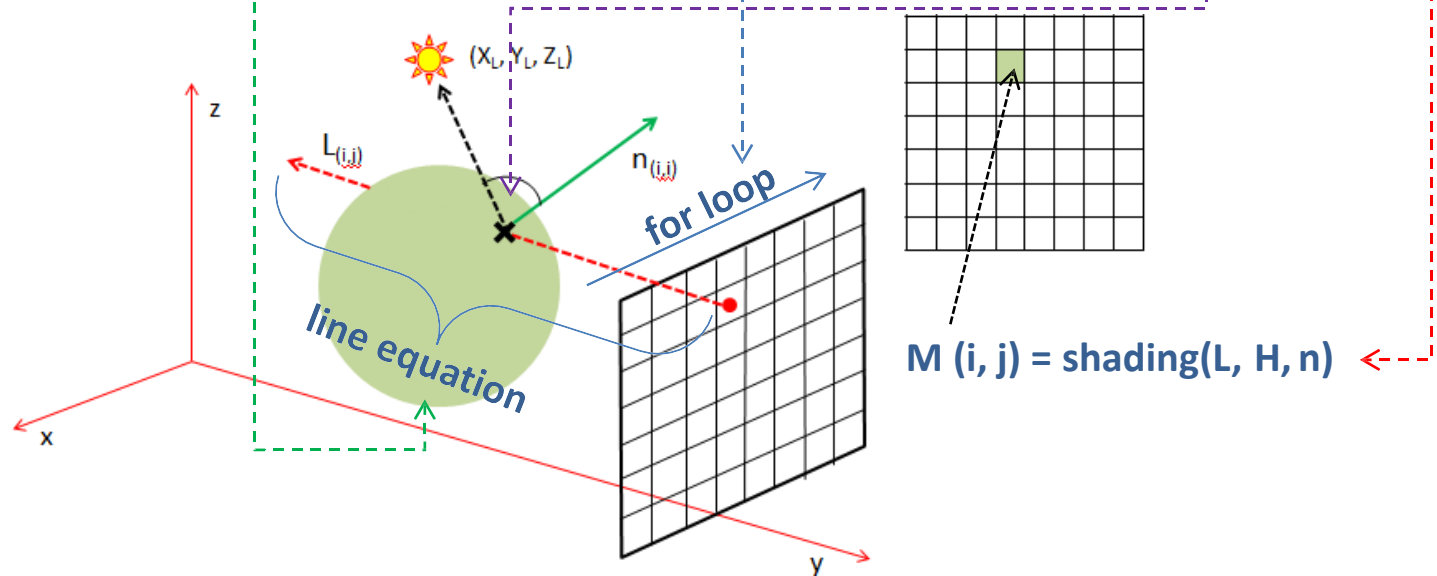
# Ray-Tracing Algorithm (2/3)

- **for each pixel do:**
  - compute viewing ray
  - find first object hit by ray and its surface normal  $\mathbf{n}$
  - set pixel color computed from hit point, light, and  $\mathbf{n}$



# Ray-Tracing Algorithm (3/3)

- **for each pixel do:**
  - compute **viewing ray**
  - find first object hit by ray and its **surface normal  $\mathbf{n}$**
  - set pixel color computed from **hit point, light, and  $\mathbf{n}$**



# Practice Problems

1. Is the projected image on the image plane in the given example perspective?
2. Consider the following setup\*:
  - Image plane: Situated at  $y = 13$ , parallel to  $ZX$  plane, (Resolution:  $11 \times 11$ ),  $M$  is the corresponding array and  $Y$ -axis goes through  $(6, 6)$ .
  - Sphere: Center at  $(0, 0, 0)$ ,  $Radius = 6$ .
  - Light: at  $(0, 15, 0)$ .

Now –

- a) Draw the ray-tracing setup showing two viewing rays (one hitting, another missing).
- b) Fill up the array (pixel) with 1 (for hitting) and 0 (for missing). Show the hitting/ missing mathematically for at least one pixel.
- c) Fill up the array (pixel) with angles between surface normal and viewing ray. Show the angle calculation for at least one pixel.

\* This problem can be helpful for understanding basic ray-tracing algorithm from the scratch.

Thank You



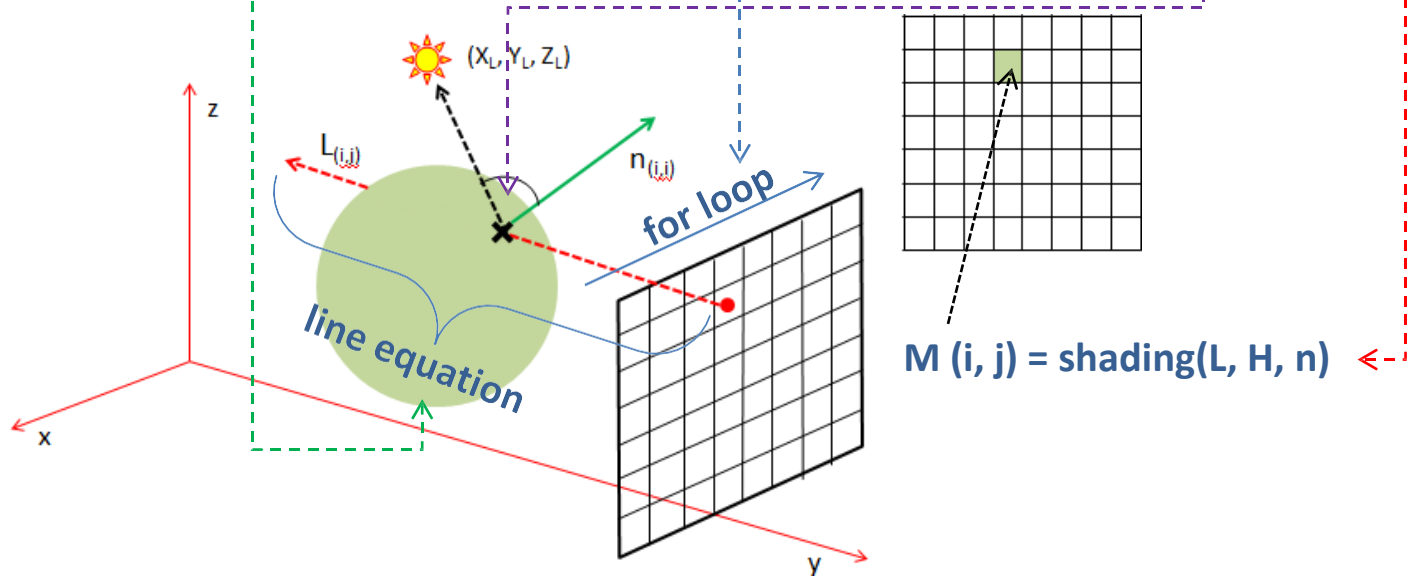
CSE4203: Computer Graphics  
Lecture – 9 (part - B)  
**Ray Tracing**

# Outline

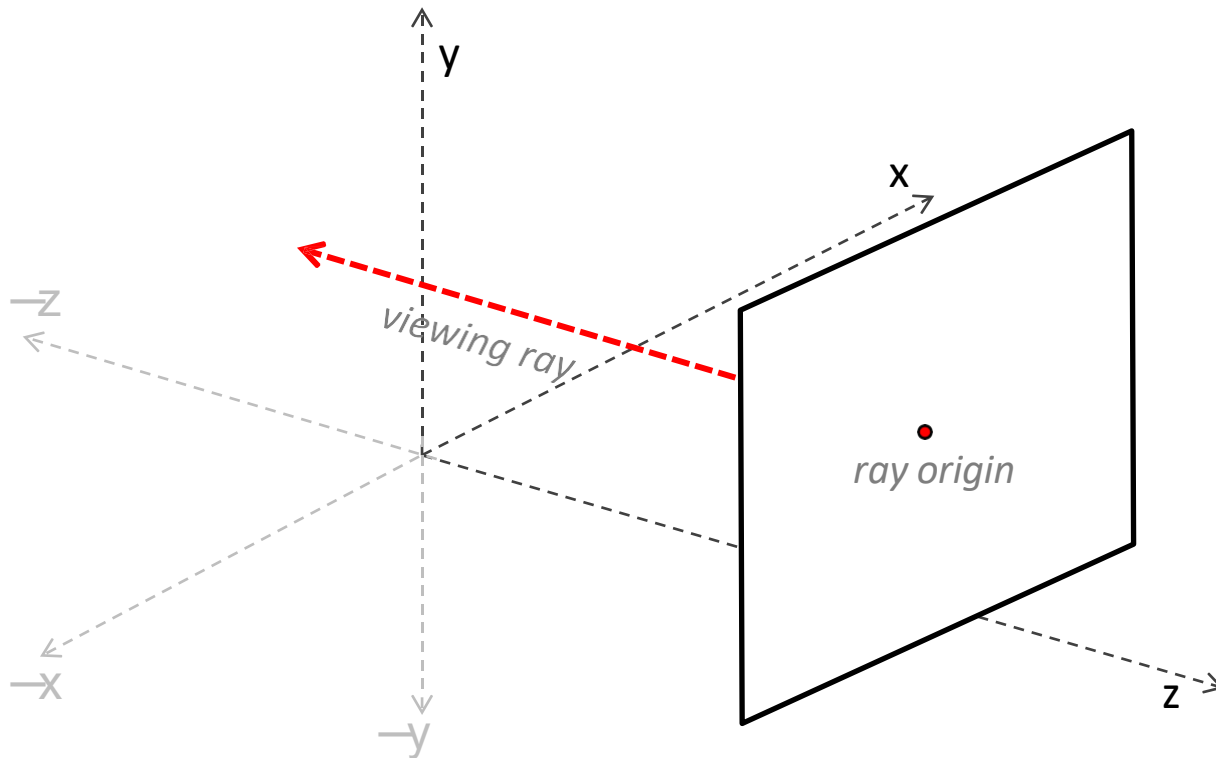
- Ray-tracing
- Camera Frame
- Image Plane and Raster Image
- Computing Viewing Rays
- Ray-sphere Intersection
- Shading

# Ray-Tracing Algorithm

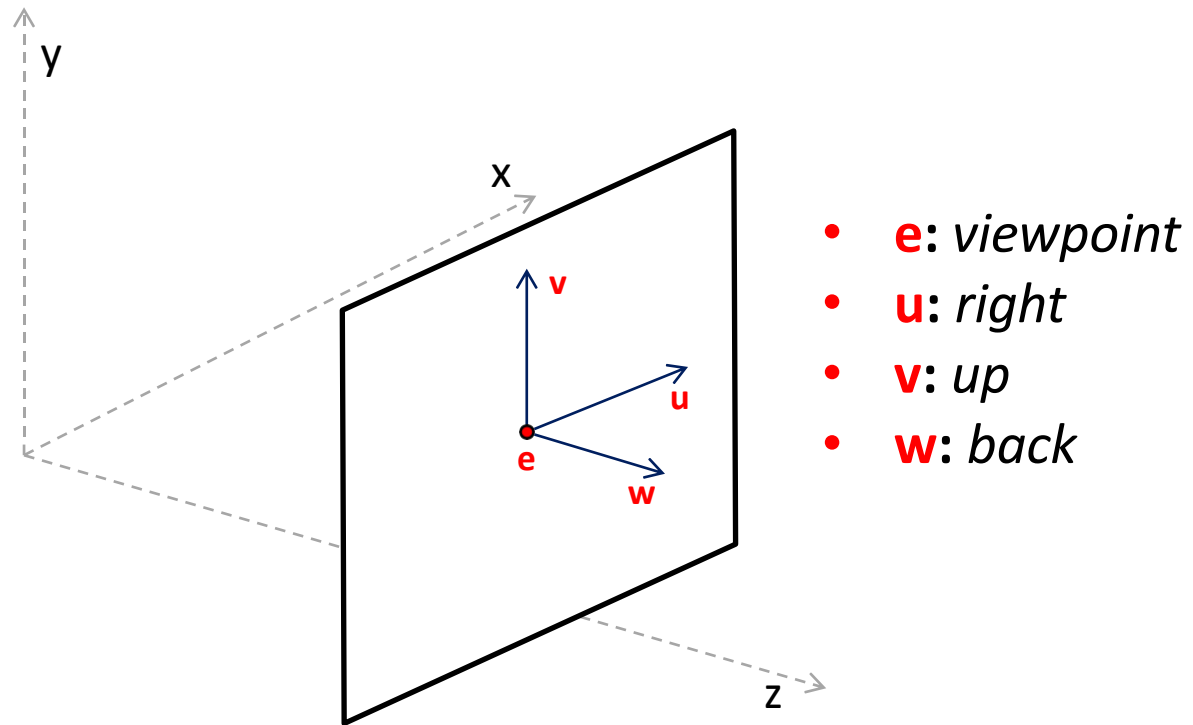
- **for each pixel do:**
  - compute **viewing ray**
  - find first object hit by ray and its **surface normal  $\mathbf{n}$**
  - set pixel color computed from **hit point, light, and  $\mathbf{n}$**



# Camera Frame (1/6)

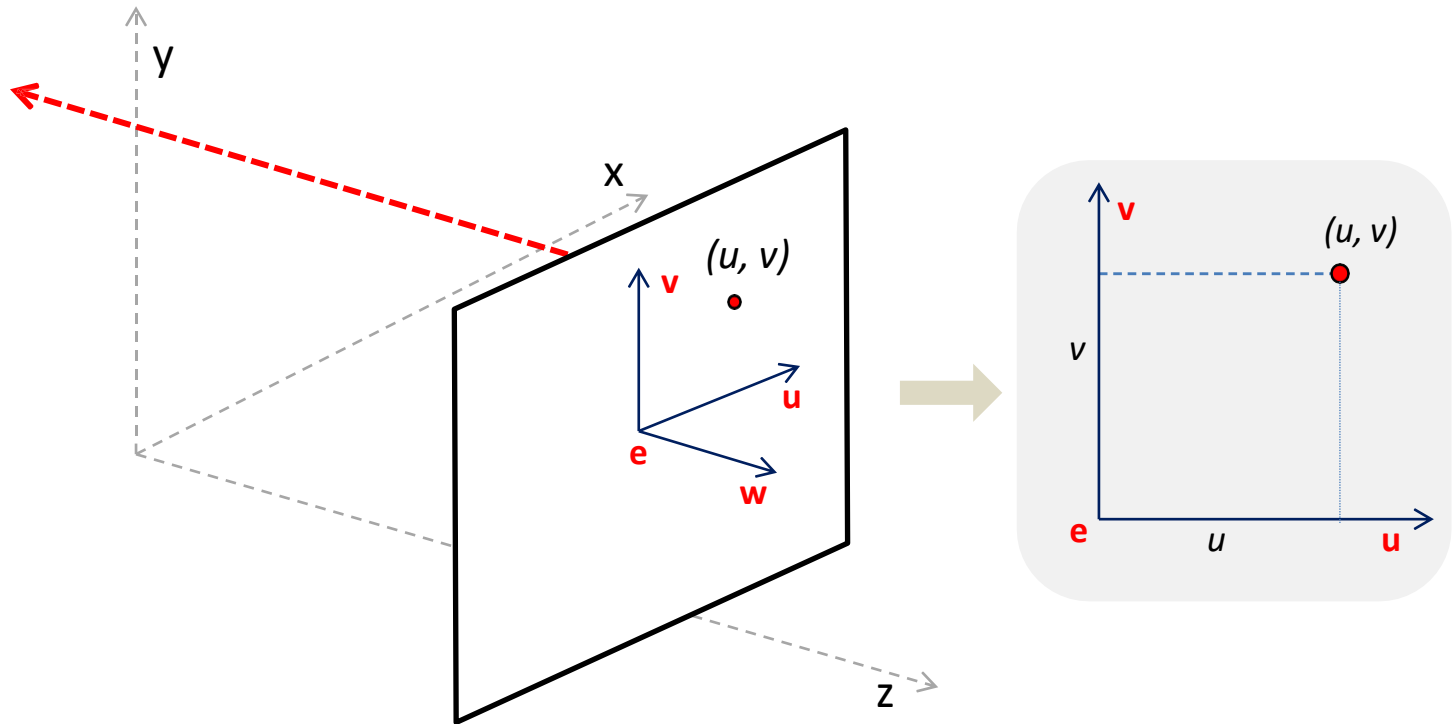


# Camera Frame (2/6)

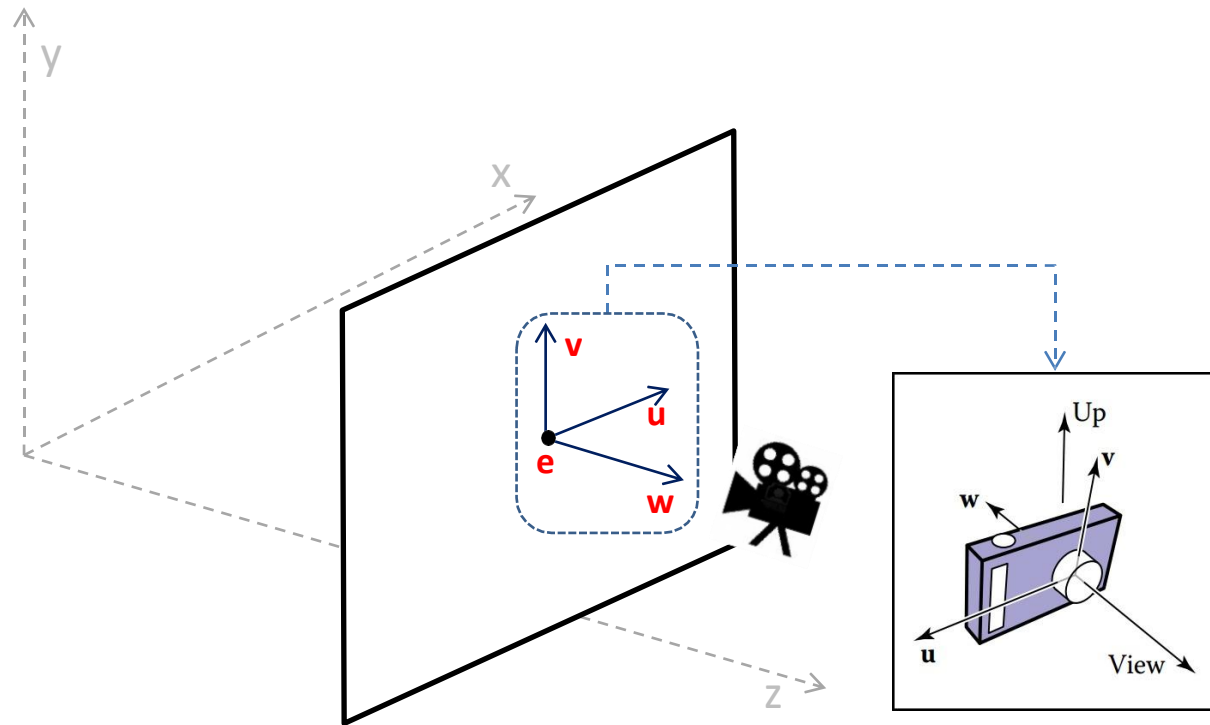


# Camera Frame (3/6)

- ray origin =  $\mathbf{e} + u \mathbf{u} + v \mathbf{v}$ 
  - ray direction =  $-\mathbf{w}$

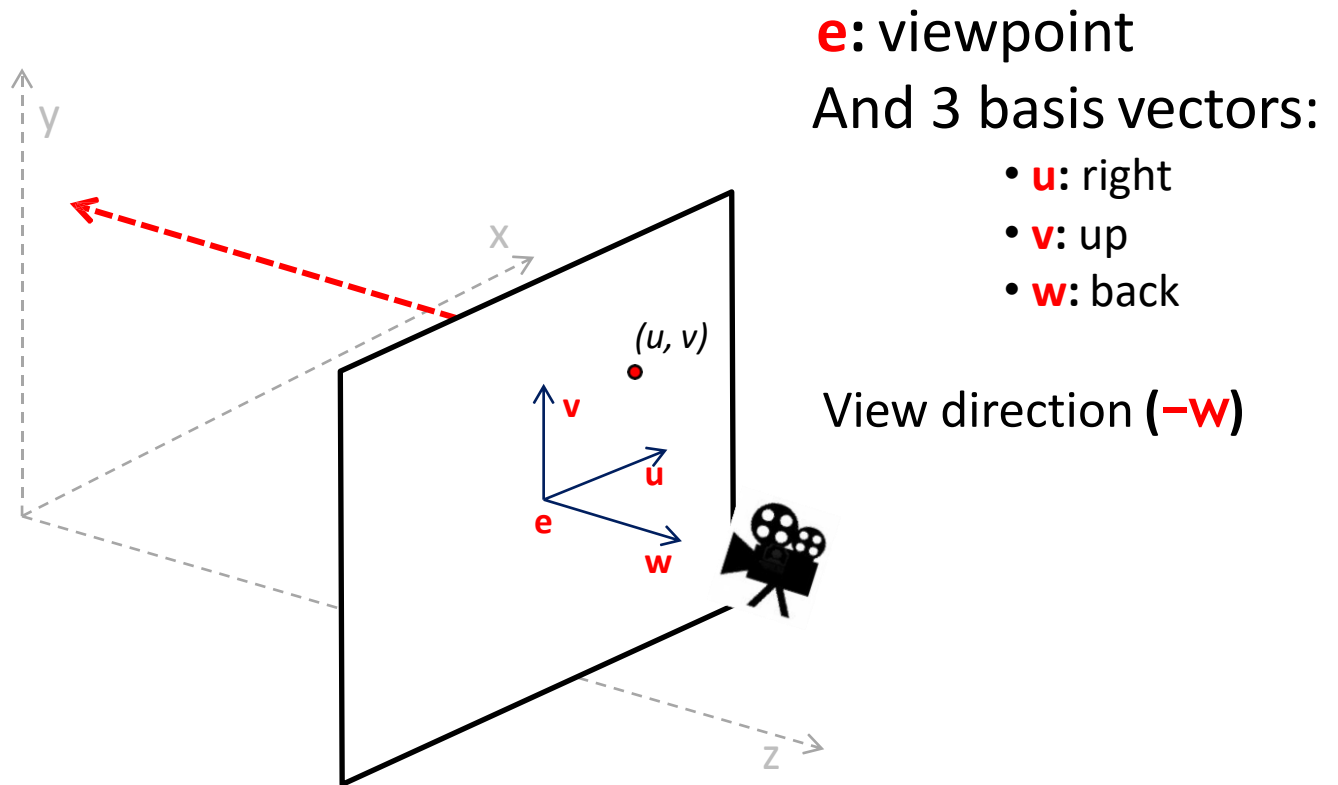


# Camera Frame (4/6)



# Camera Frame (5/6)

Camera frame: (Camera coordinate)

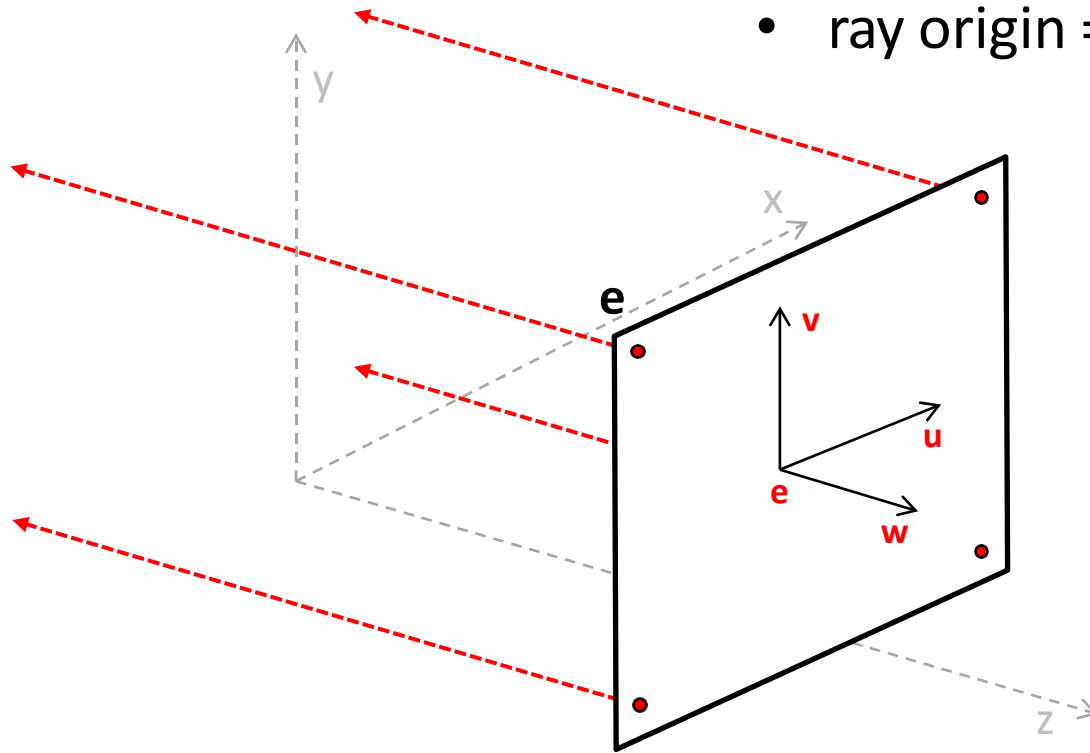




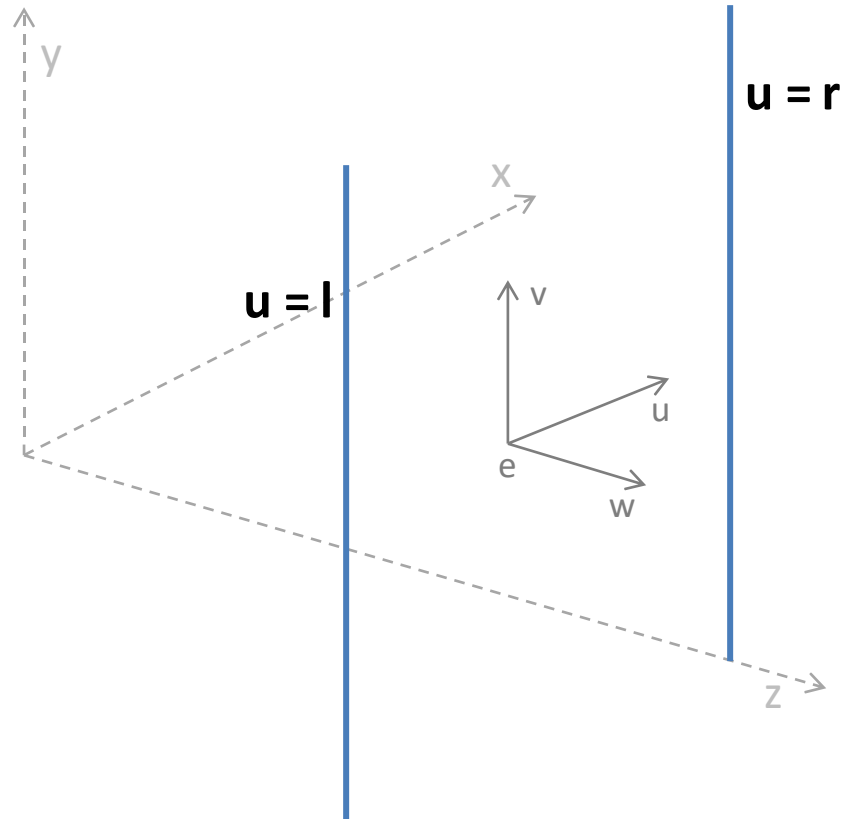
# Camera Frame (6/6)

Orthographic:

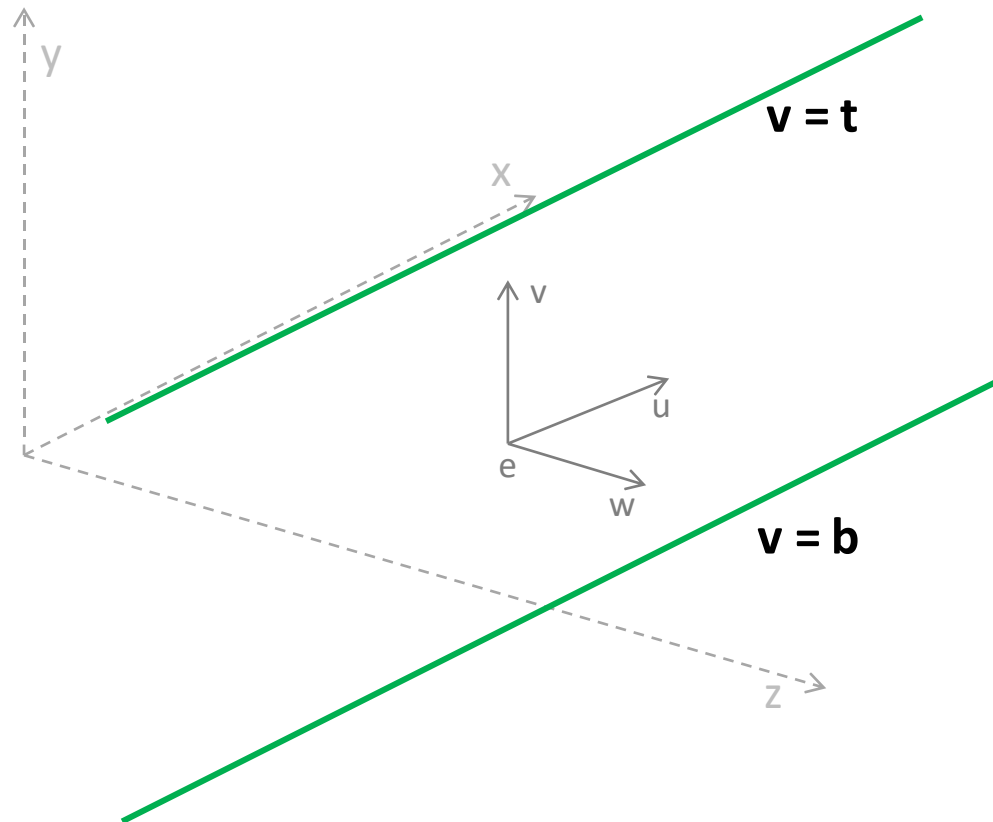
- ray direction =  $-\mathbf{w}$
- ray origin =  $\mathbf{e} + u \mathbf{u} + v \mathbf{v}$



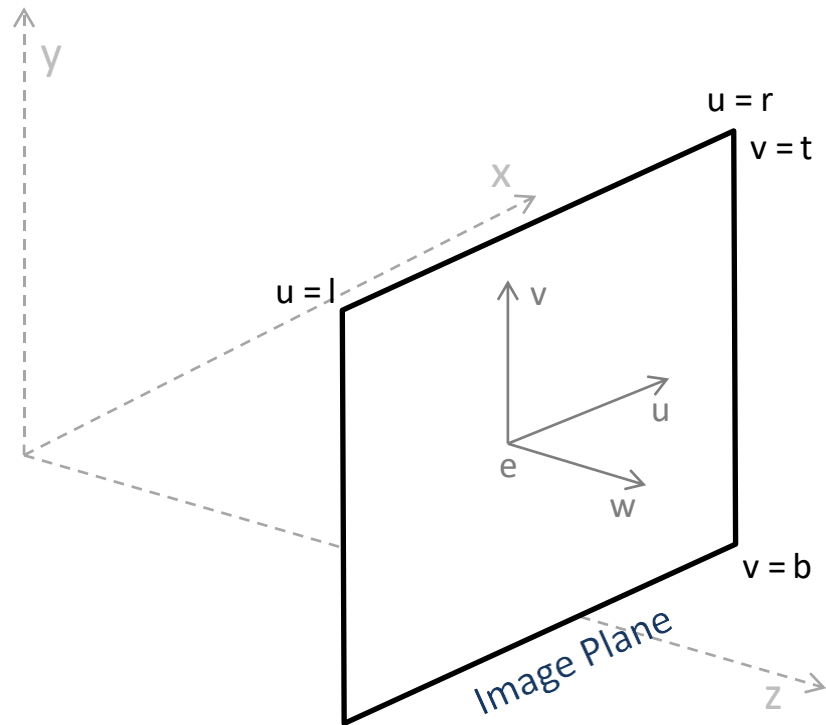
# Image Plane (1/4)



# Image Plane (2/4)

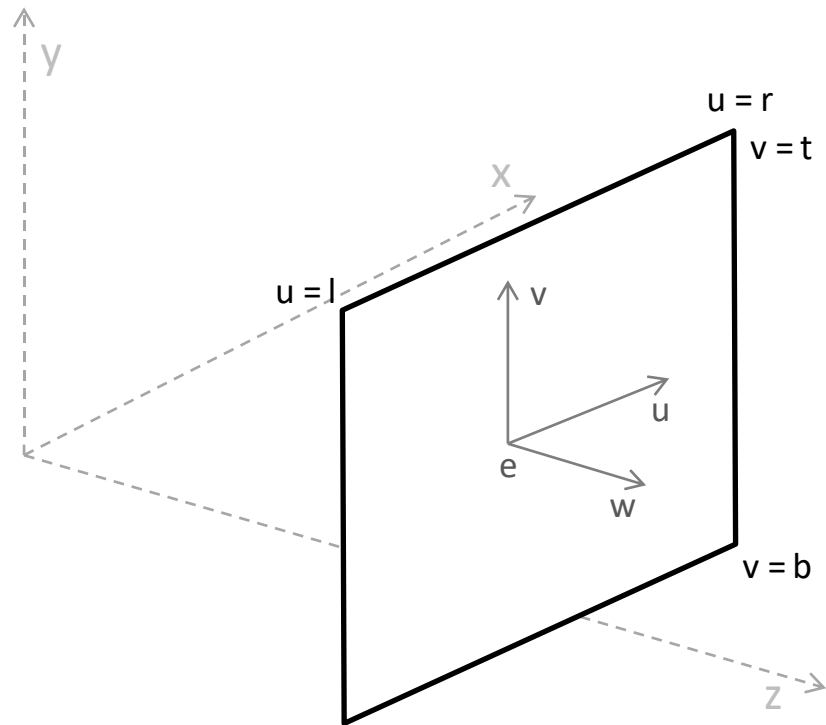


# Image Plane (3/4)

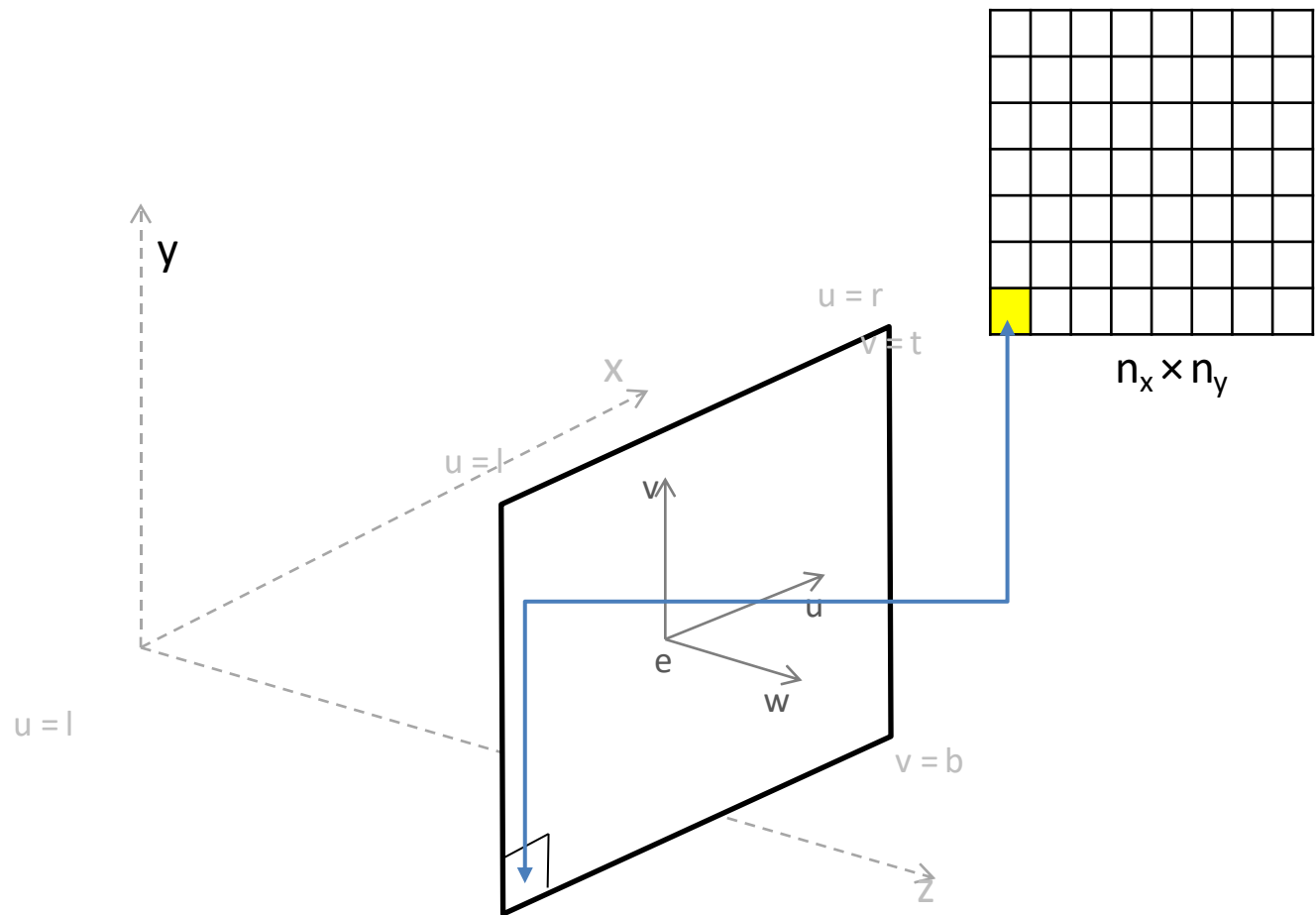


# Image Plane (4/4)

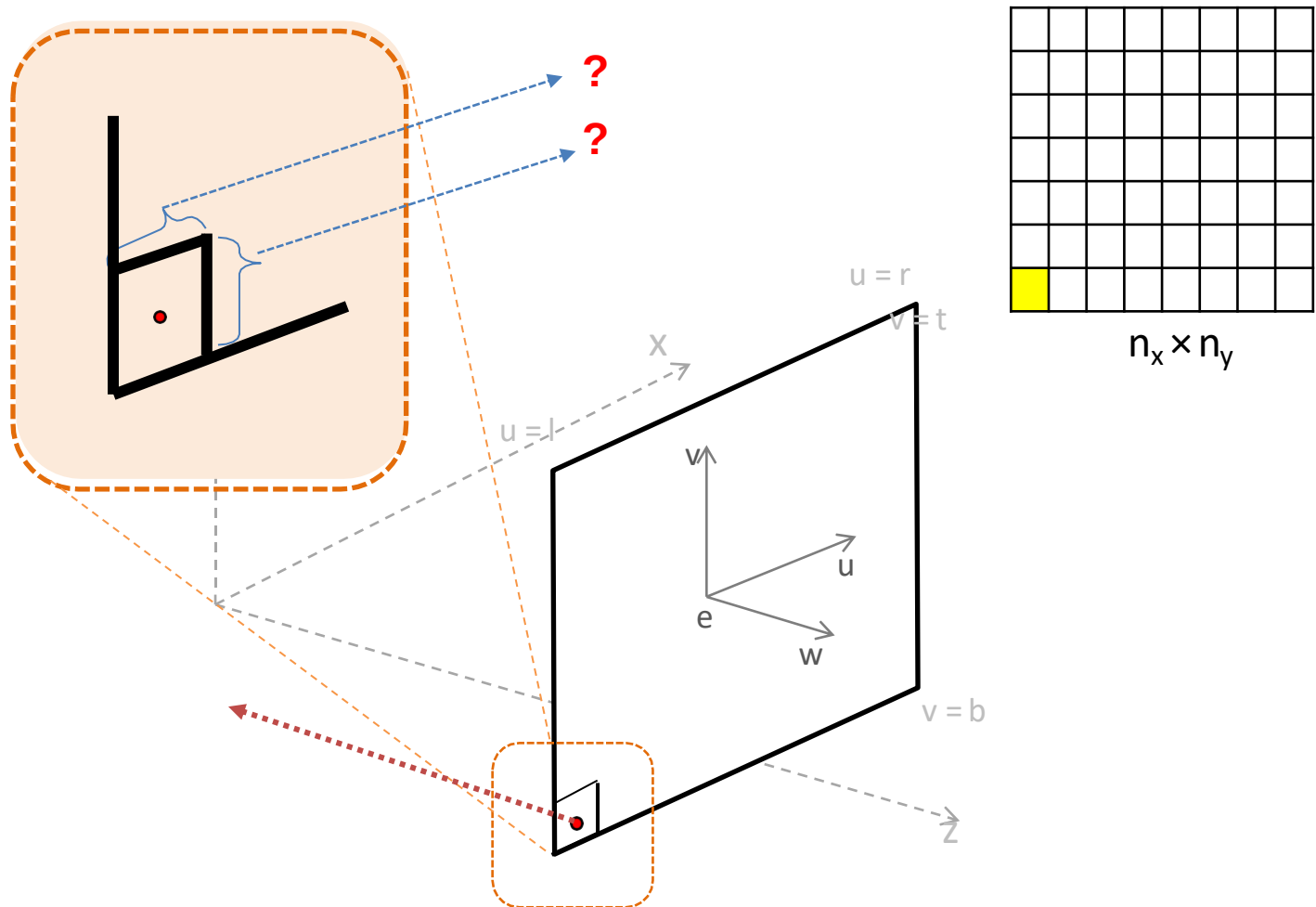
Q: determine the area of the image plane in terms of  $l$ ,  $r$ ,  $t$  and  $b$ .



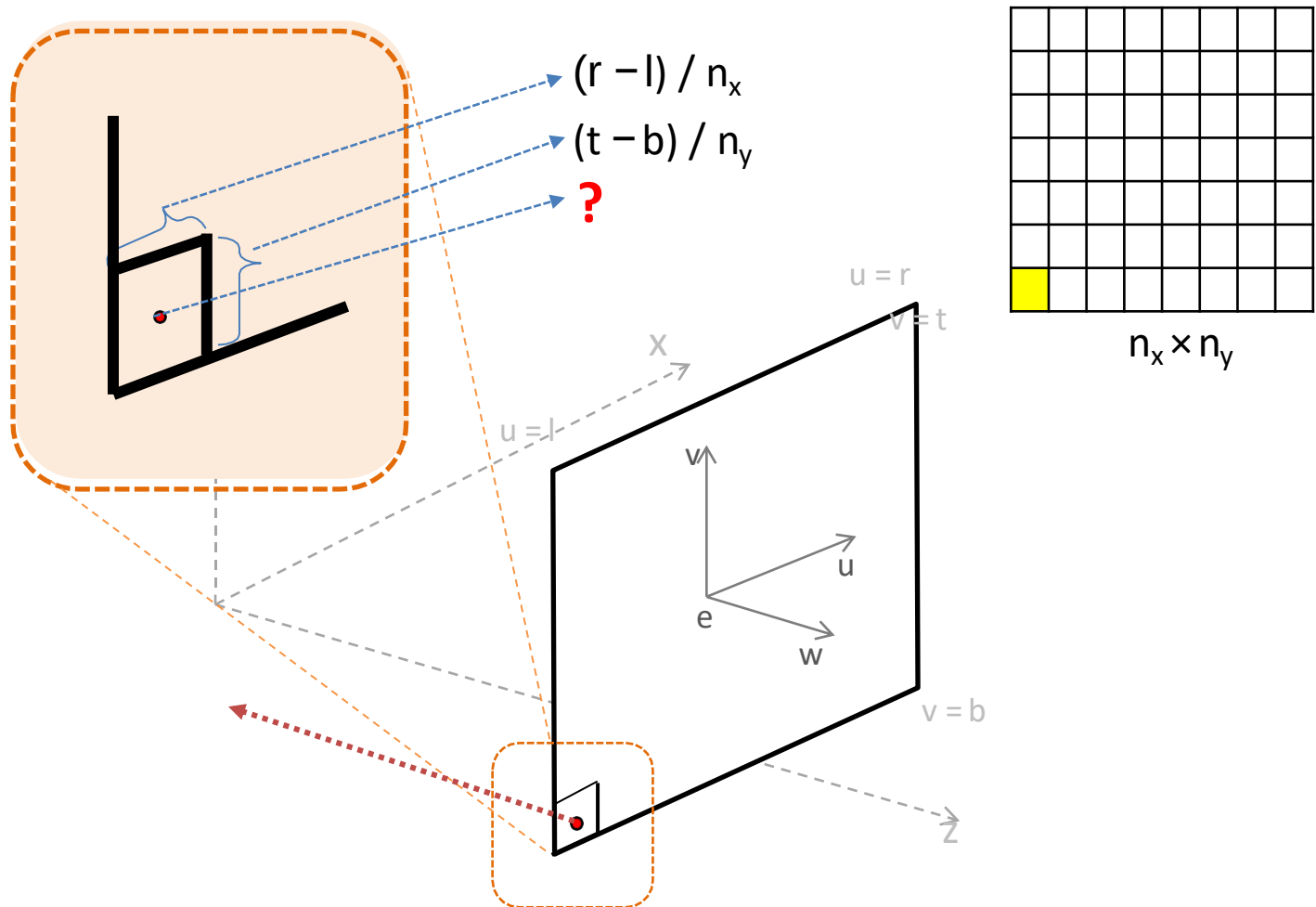
# *Raster Image $\leftrightarrow$ Image Plane (1/8)*



# *Raster Image $\leftrightarrow$ Image Plane (2/8)*

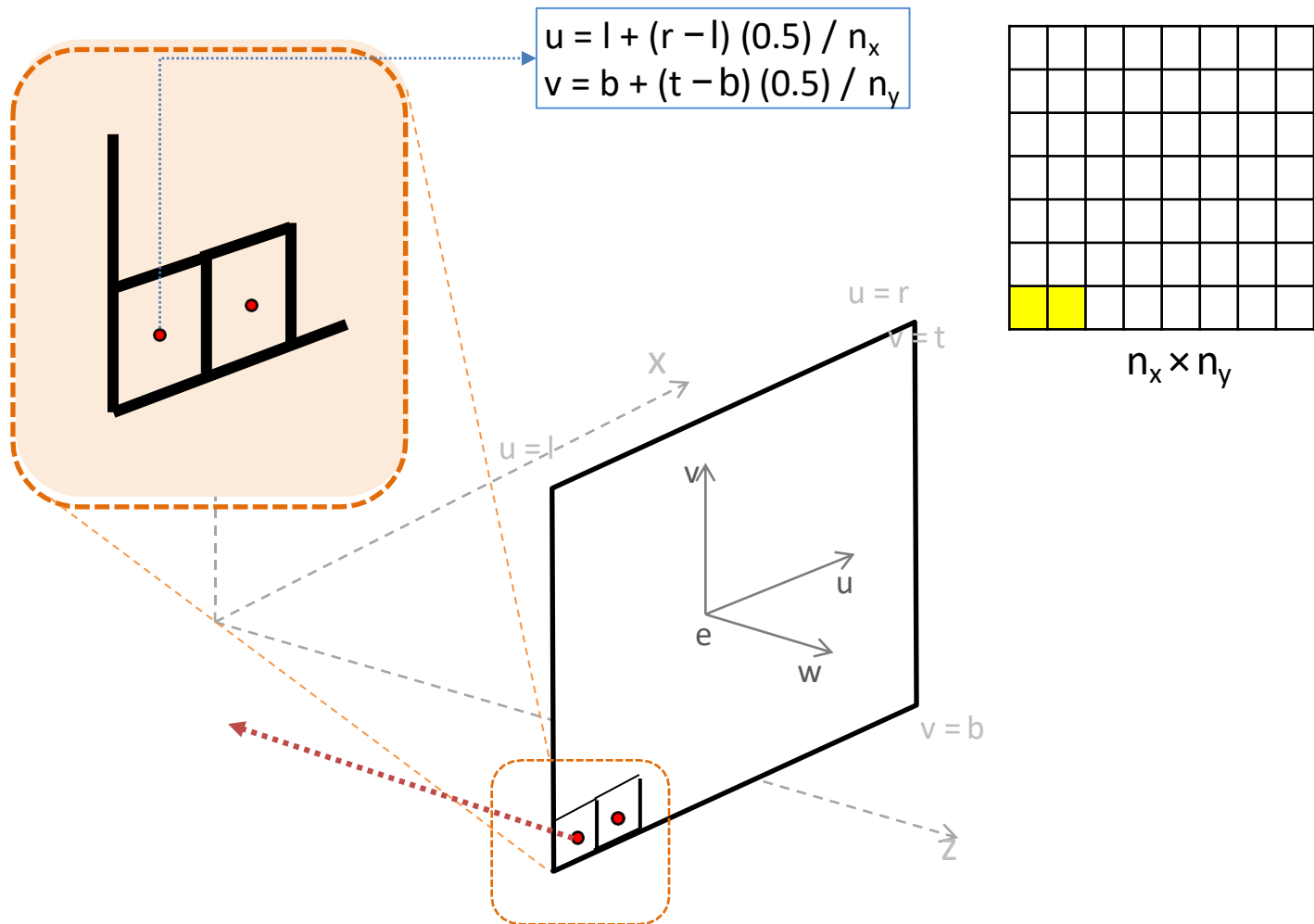


# *Raster Image $\leftrightarrow$ Image Plane (3/8)*

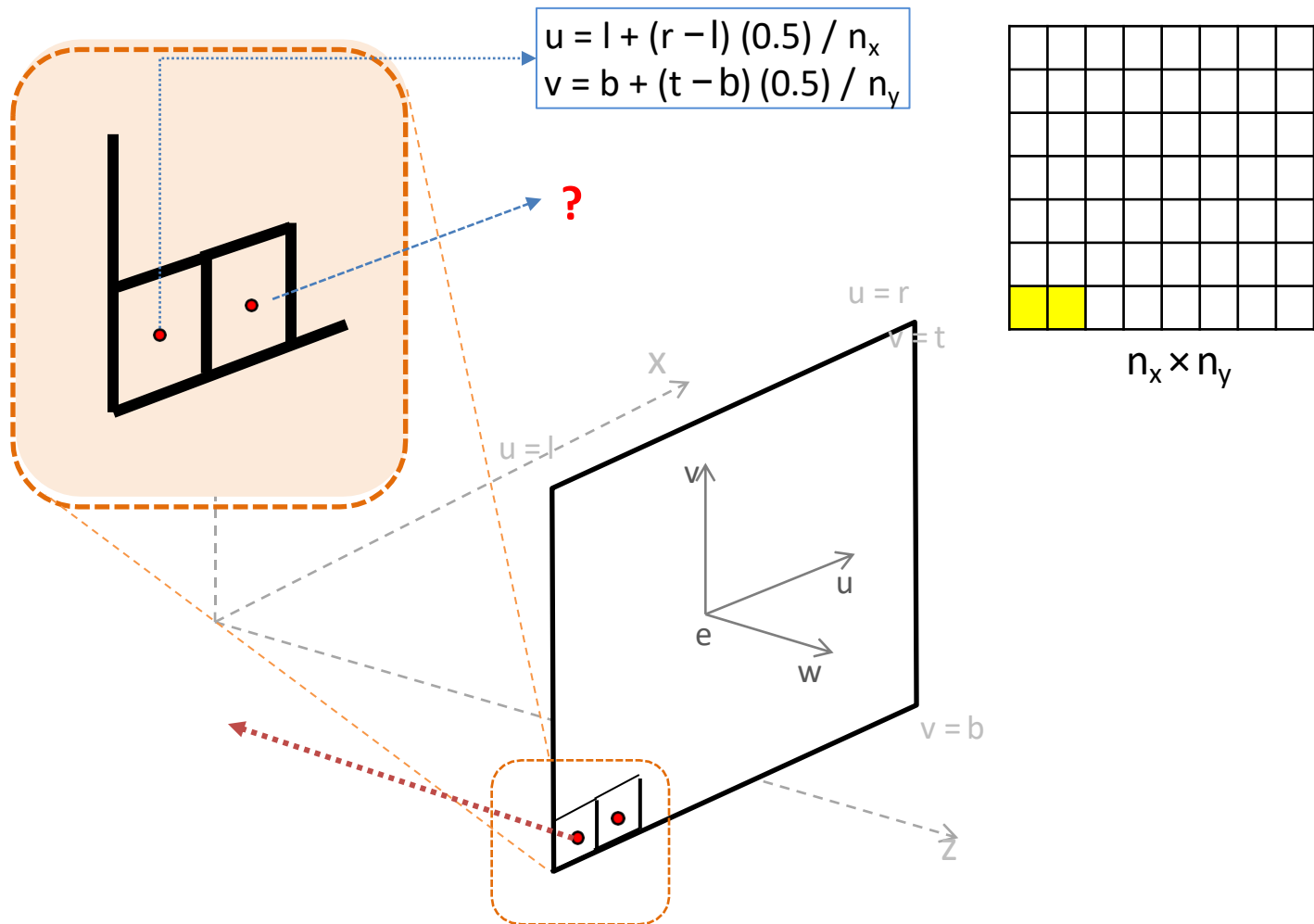




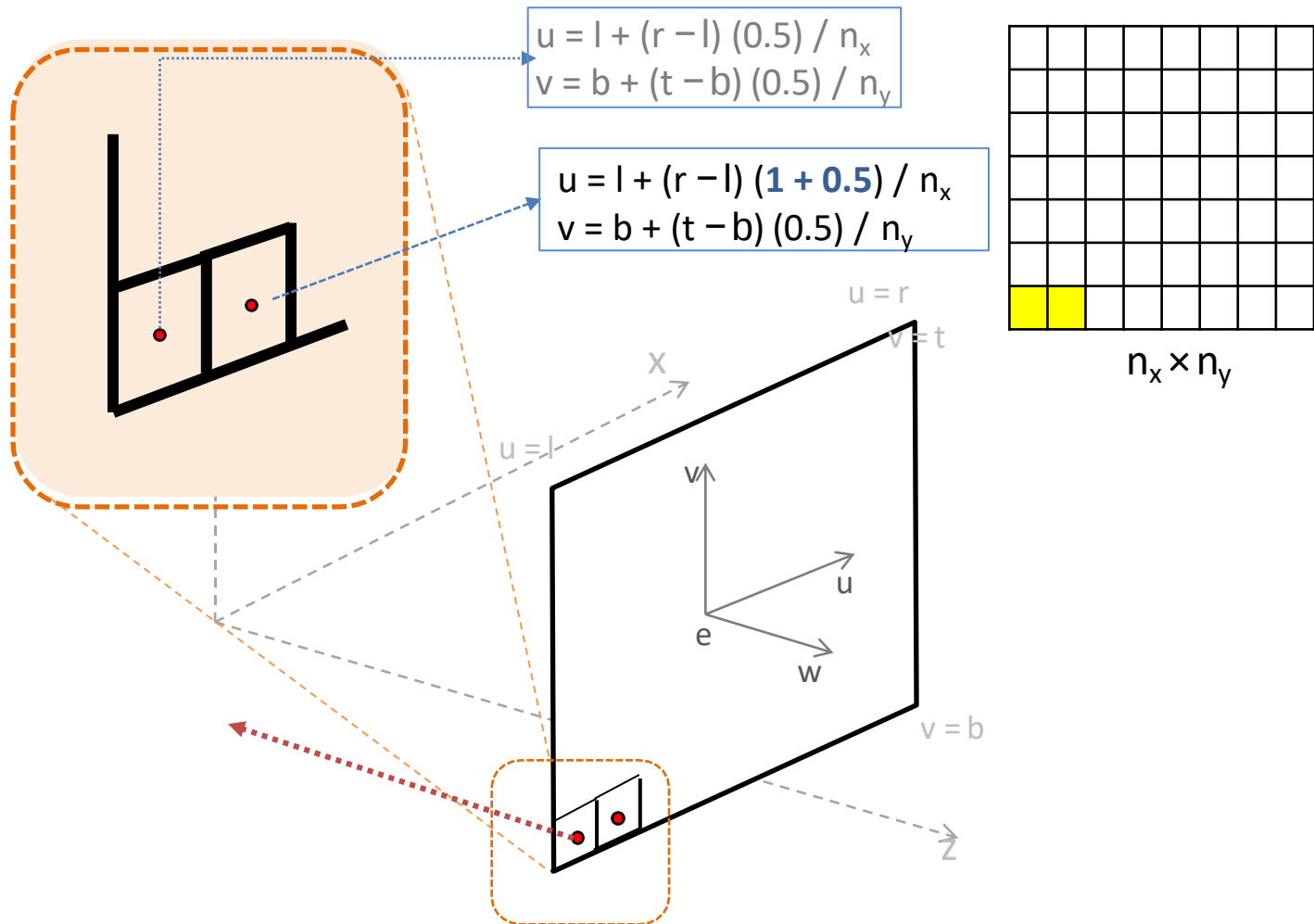
# *Raster Image $\leftrightarrow$ Image Plane (4/8)*



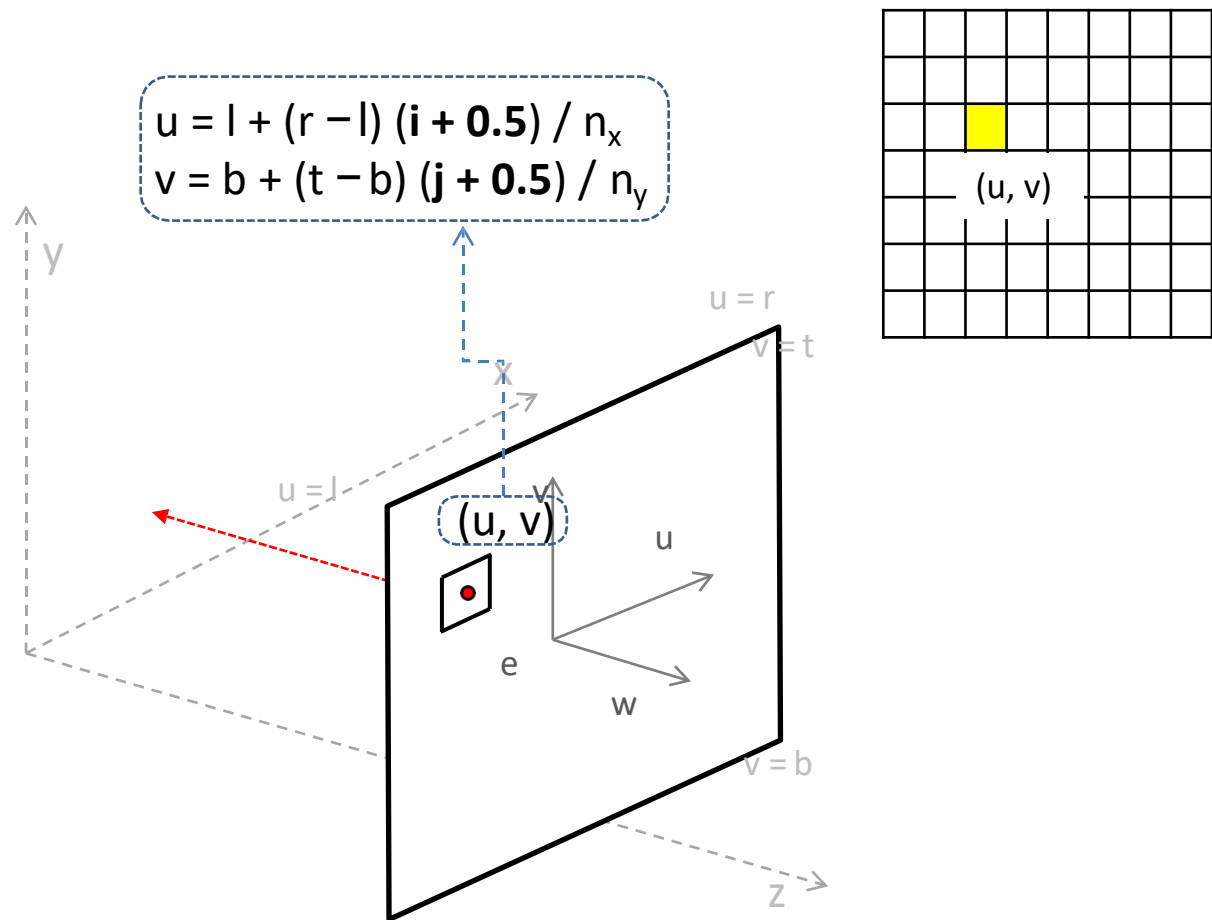
# Raster Image $\leftrightarrow$ Image Plane (5/8)



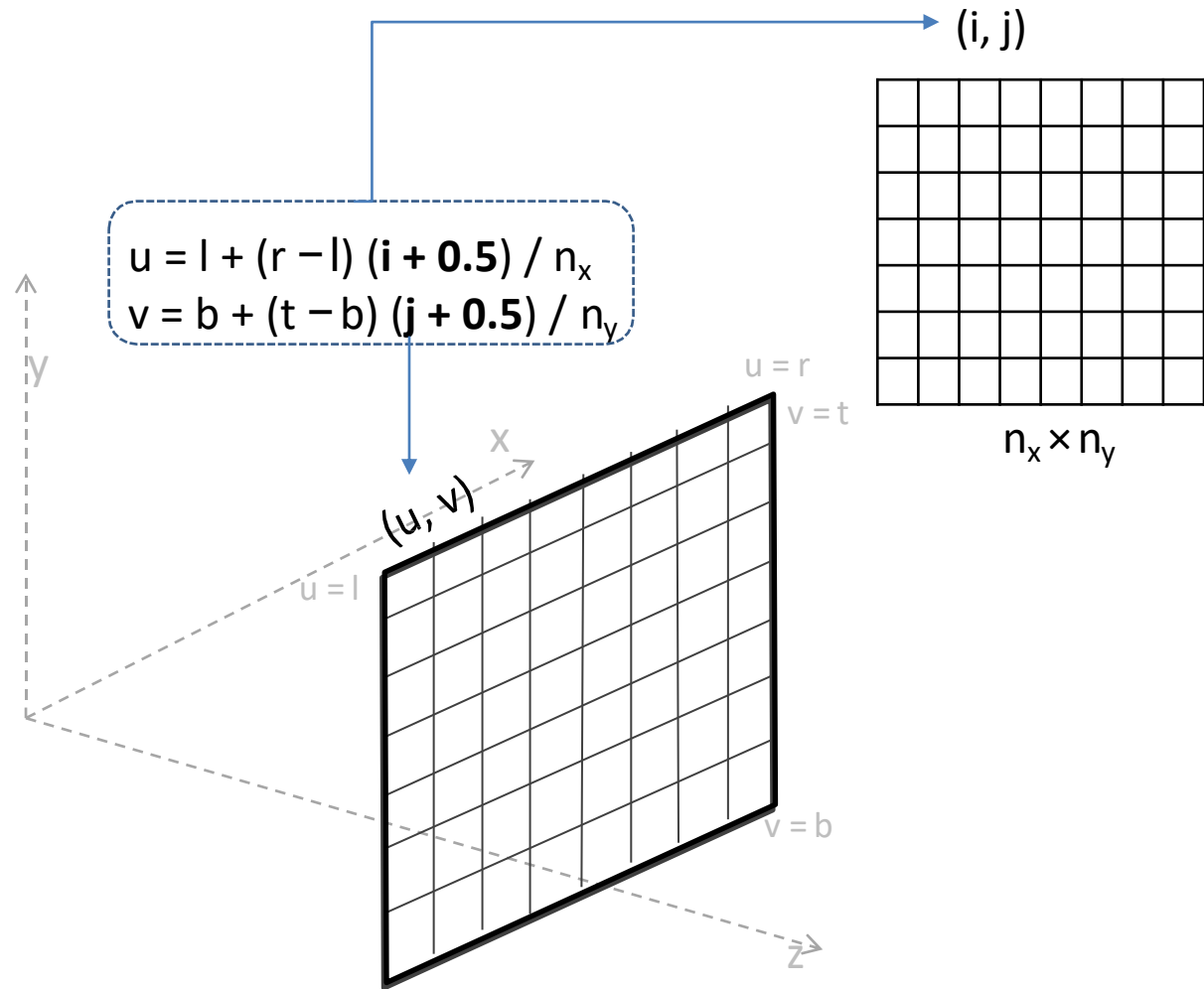
# Raster Image $\leftrightarrow$ Image Plane (6/8)



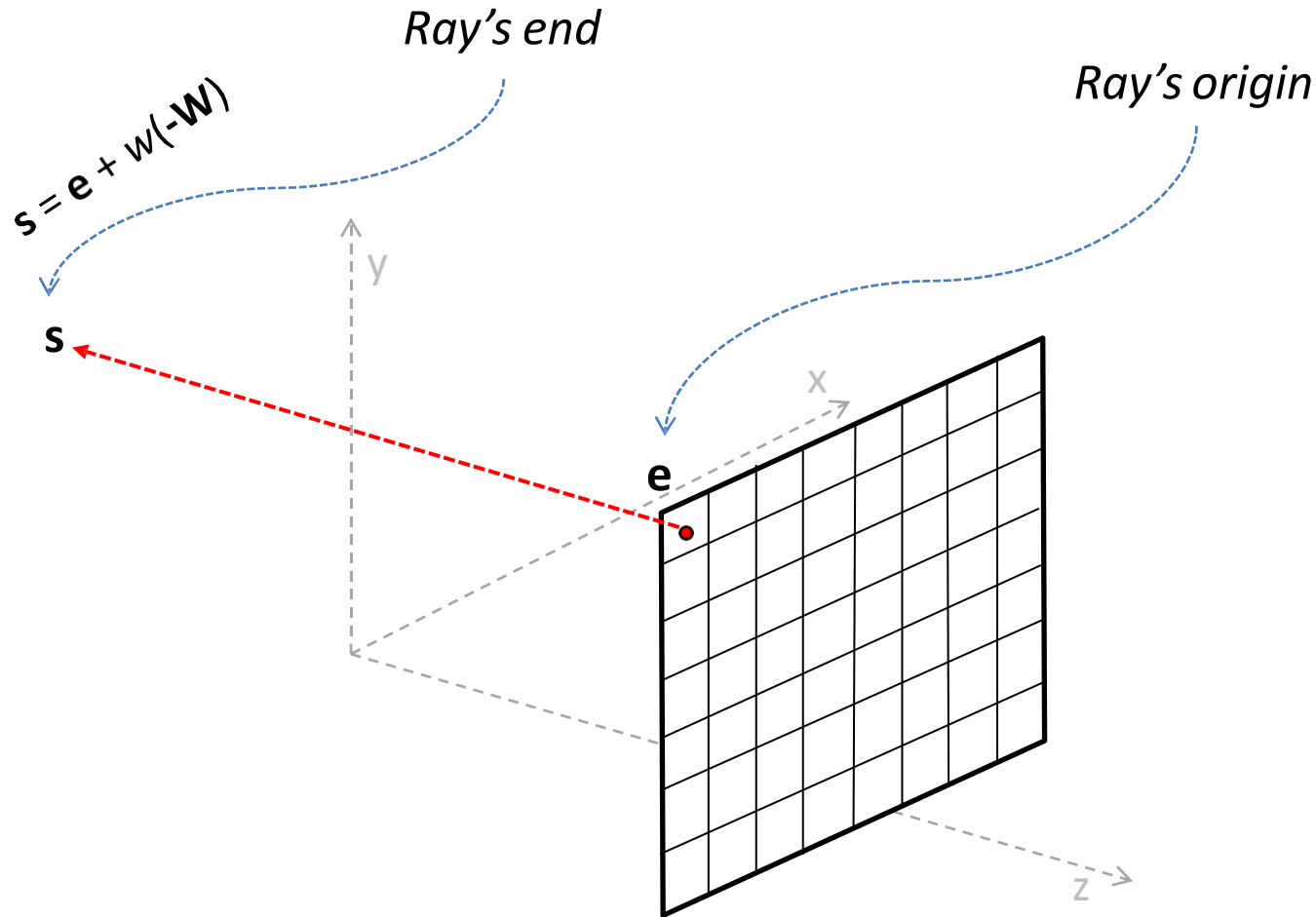
# *Raster Image $\leftrightarrow$ Image Plane (7/8)*



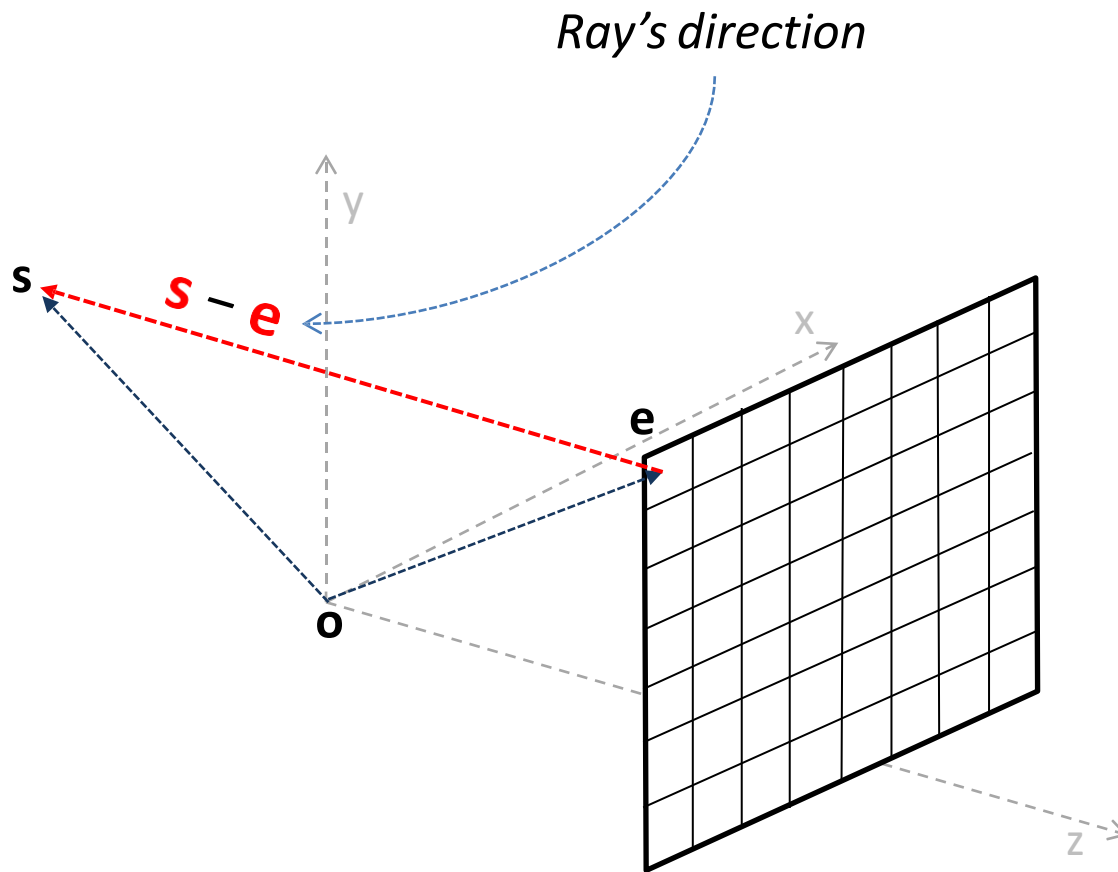
# *Raster Image $\leftrightarrow$ Image Plane (8/8)*



# Computing Viewing Rays (1/4)



# Computing Viewing Rays (2/4)

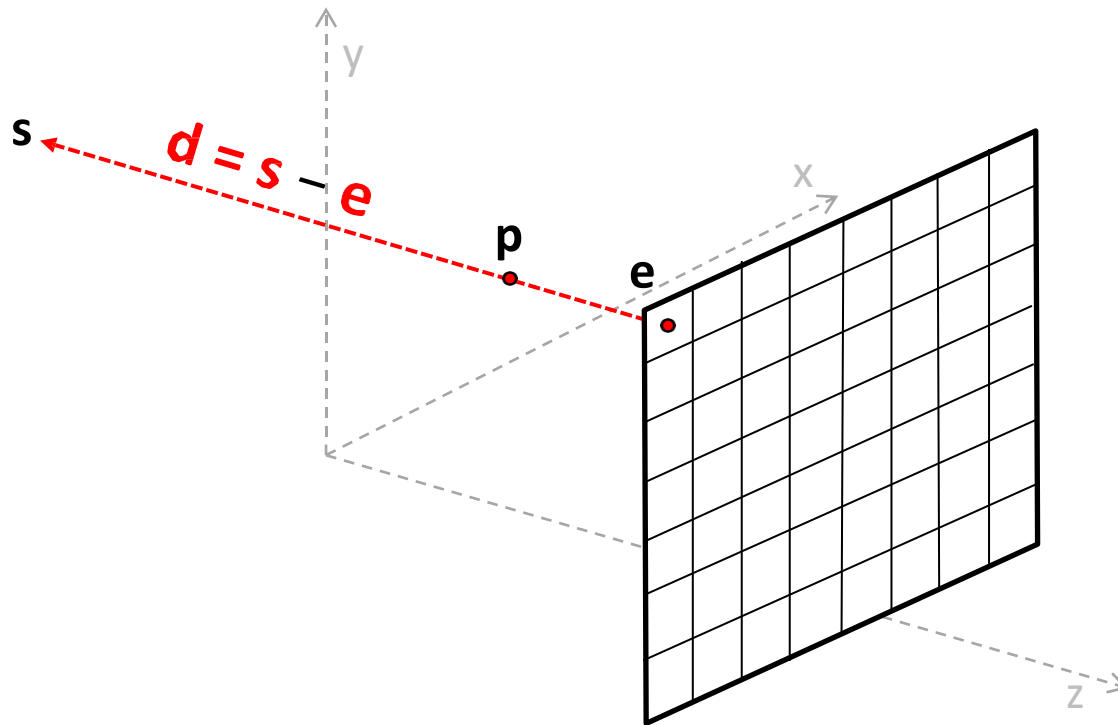


# Computing Viewing Rays (3/4)

$$\begin{aligned}\mathbf{p} &= \mathbf{e} + t(\mathbf{s} - \mathbf{e}) \\ &= \mathbf{e} + t\mathbf{d}\end{aligned}$$

- Advancing from  $\mathbf{e}$  along the vector  $(\mathbf{s} - \mathbf{e})$

- With a fractional distance  $t$  to find the point  $\mathbf{p}$

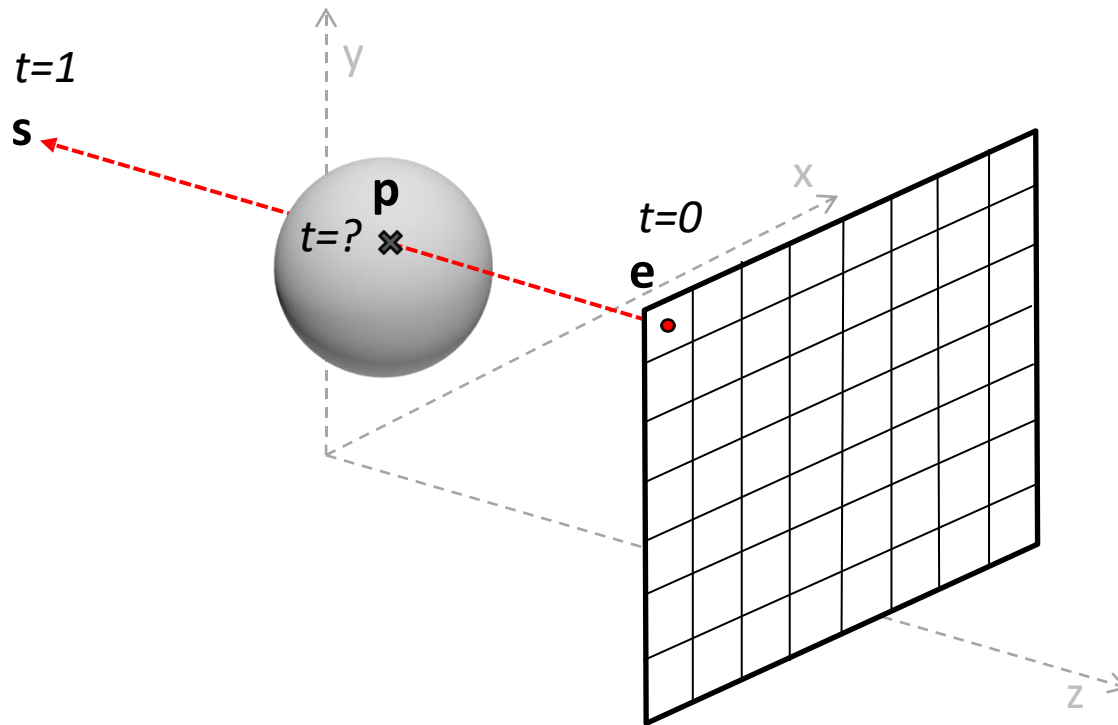




# Computing Viewing Rays (4/4)

$$\begin{aligned}\mathbf{p} &= \mathbf{e} + t(\mathbf{s} - \mathbf{e}) \\ &= \mathbf{e} + t\mathbf{d}\end{aligned}$$

- We can use  $t$  to determine the intersection point  $\mathbf{p}$

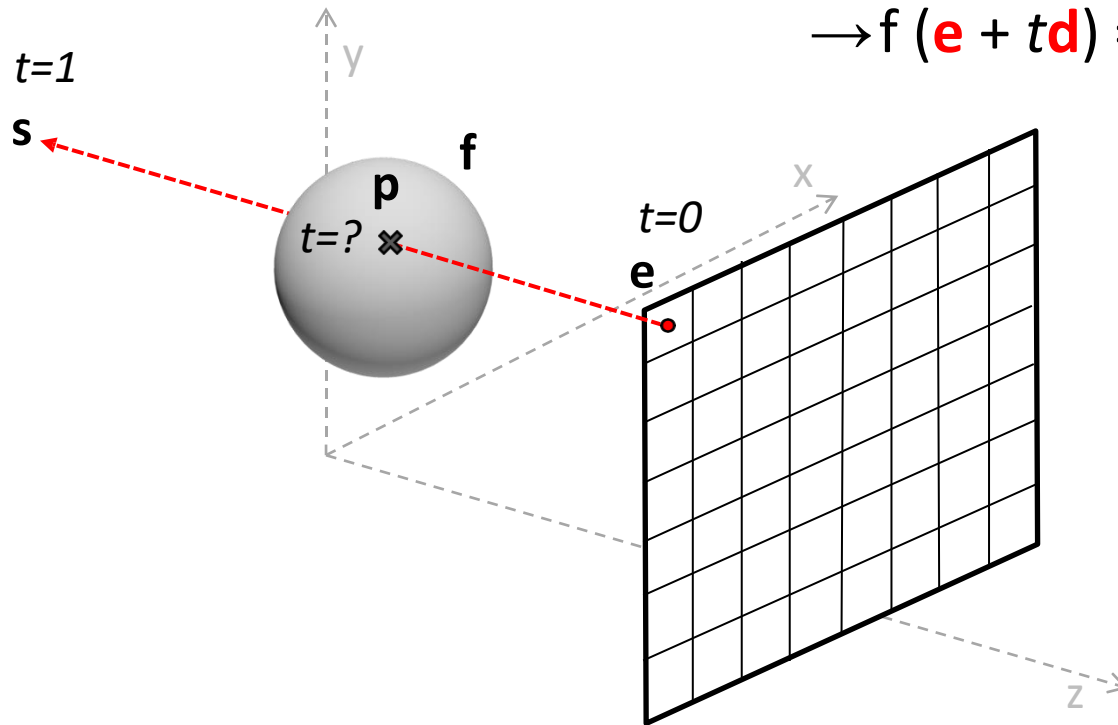


# Ray - Sphere Intersection (1/8)

We have,  $\mathbf{p} = \mathbf{e} + t(\mathbf{s} - \mathbf{e}) = \mathbf{e} + t\mathbf{d}$

$$\rightarrow f(\mathbf{p}) = 0$$

$$\rightarrow f(\mathbf{e} + t\mathbf{d}) = 0$$

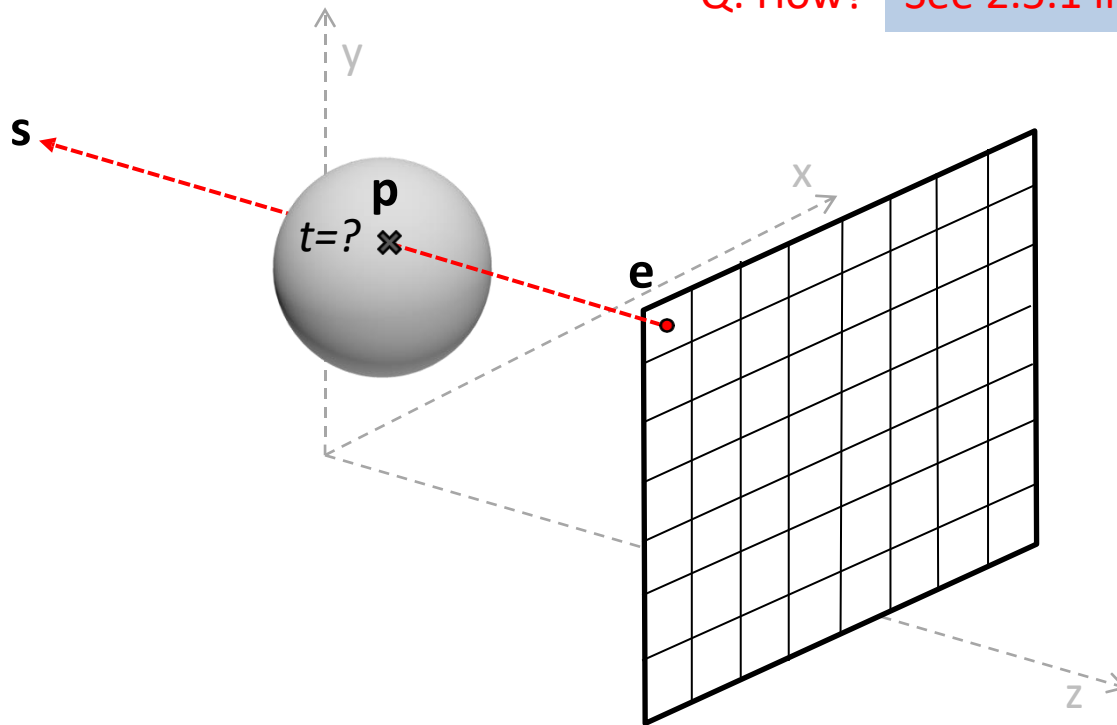


# Ray - Sphere Intersection (2/8)

$$\rightarrow (x - x_c)^2 + (y - y_c)^2 + (z - z_c)^2 - R^2 = 0$$

$$\rightarrow (\mathbf{p} - \mathbf{c}) \cdot (\mathbf{p} - \mathbf{c}) - R^2 = 0$$

Q: How? See 2.5.1 in Book



# Ray - Sphere Intersection (3/8)

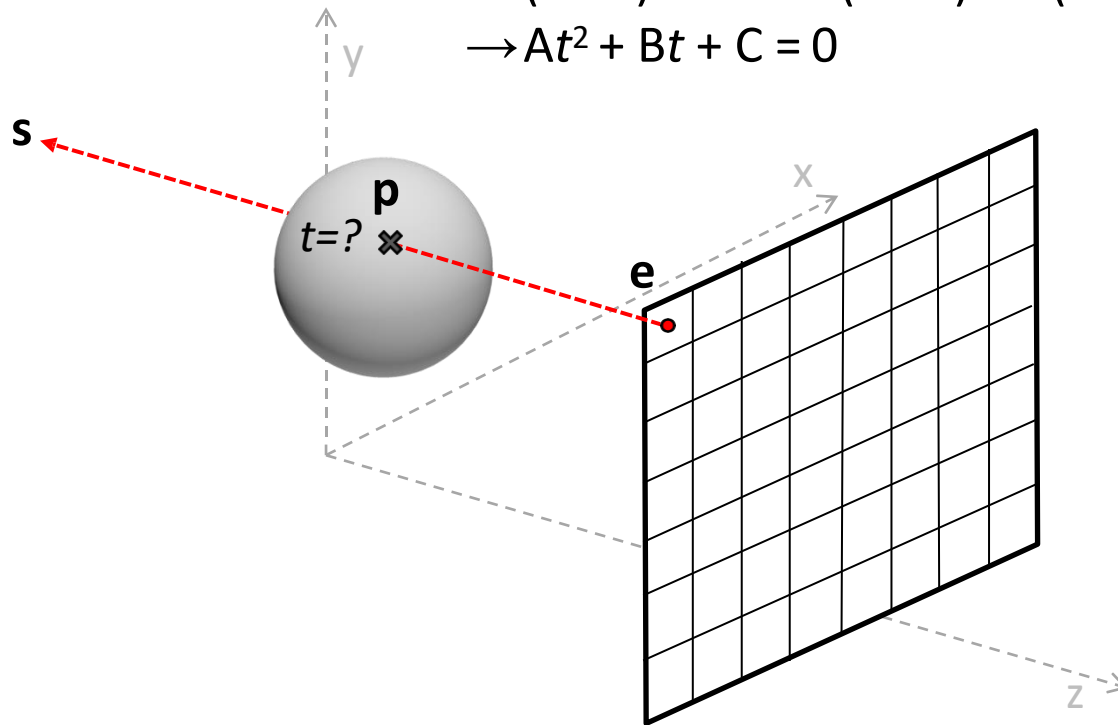
$$\rightarrow (x - x_c)^2 + (y - y_c)^2 + (z - z_c)^2 - R^2 = 0$$

$$\rightarrow (\mathbf{p} - \mathbf{c}) \cdot (\mathbf{p} - \mathbf{c}) - R^2 = 0$$

$$\rightarrow (\mathbf{e} + t\mathbf{d} - \mathbf{c}) \cdot (\mathbf{e} + t\mathbf{d} - \mathbf{c}) - R^2 = 0$$

$$\rightarrow (\mathbf{d} \cdot \mathbf{d}) t^2 + 2\mathbf{d} \cdot (\mathbf{e} - \mathbf{c}) t + (\mathbf{e} - \mathbf{c}) \cdot (\mathbf{e} - \mathbf{c}) - R^2 = 0$$

$$\rightarrow At^2 + Bt + C = 0$$



# Ray - Sphere Intersection (4/8)

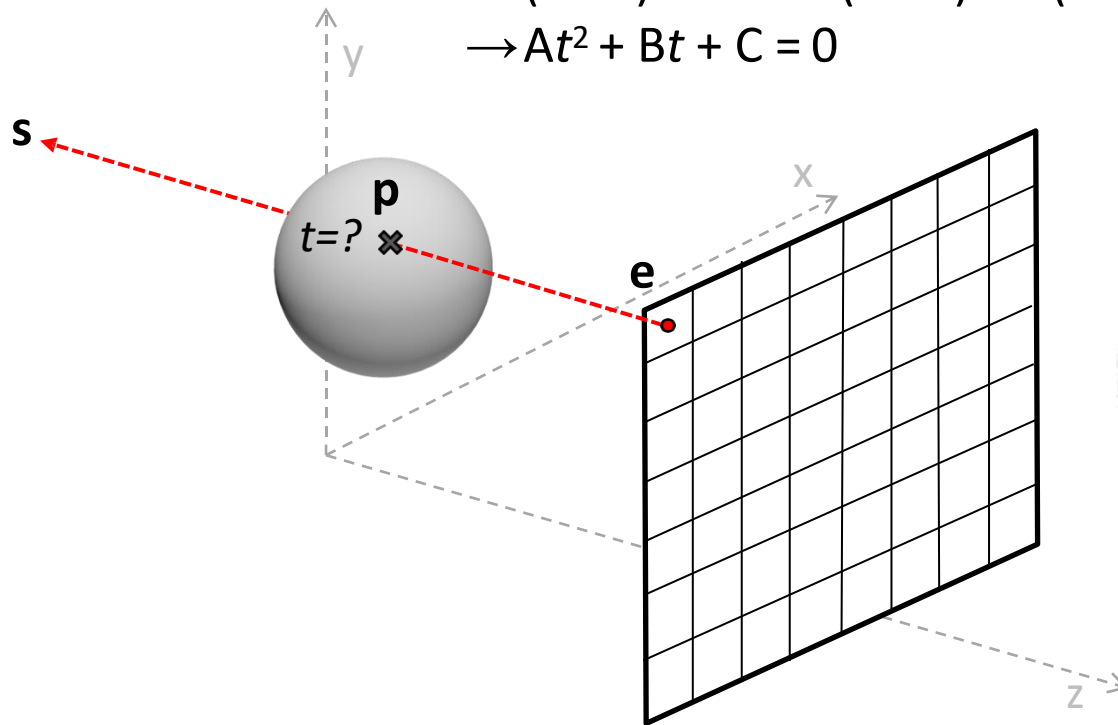
$$\rightarrow (x - x_c)^2 + (y - y_c)^2 + (z - z_c)^2 - R^2 = 0$$

$$\rightarrow (\mathbf{p} - \mathbf{c}) \cdot (\mathbf{p} - \mathbf{c}) - R^2 = 0$$

$$\rightarrow (\mathbf{e} + t\mathbf{d} - \mathbf{c}) \cdot (\mathbf{e} + t\mathbf{d} - \mathbf{c}) - R^2 = 0$$

$$\rightarrow (\mathbf{d} \cdot \mathbf{d}) t^2 + 2\mathbf{d} \cdot (\mathbf{e} - \mathbf{c}) t + (\mathbf{e} - \mathbf{c}) \cdot (\mathbf{e} - \mathbf{c}) - R^2 = 0$$

$$\rightarrow At^2 + Bt + C = 0$$



$$t = \frac{-B \pm \sqrt{B^2 - 4AC}}{2A}$$

# Ray - Sphere Intersection (5/8)

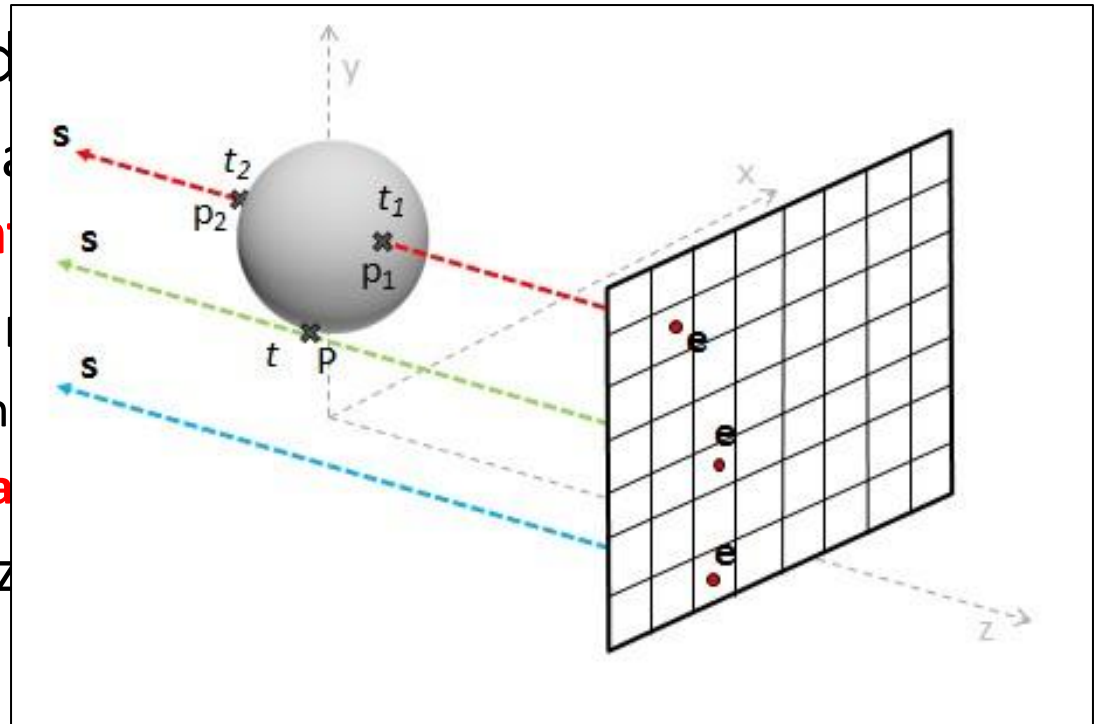
$$t = \frac{-B \pm \sqrt{B^2 - 4AC}}{2A}$$

- $B^2 - 4AC$ , is called the discriminant and if it is –
  - **negative**: its square root is imaginary and the line and sphere **do not intersect**.
  - **positive**: there are two solutions –
    - one solution where the ray **enters** the sphere.
    - one where it **leaves**.
  - **zero**: the ray grazes the sphere, touching it at exactly **one point**.

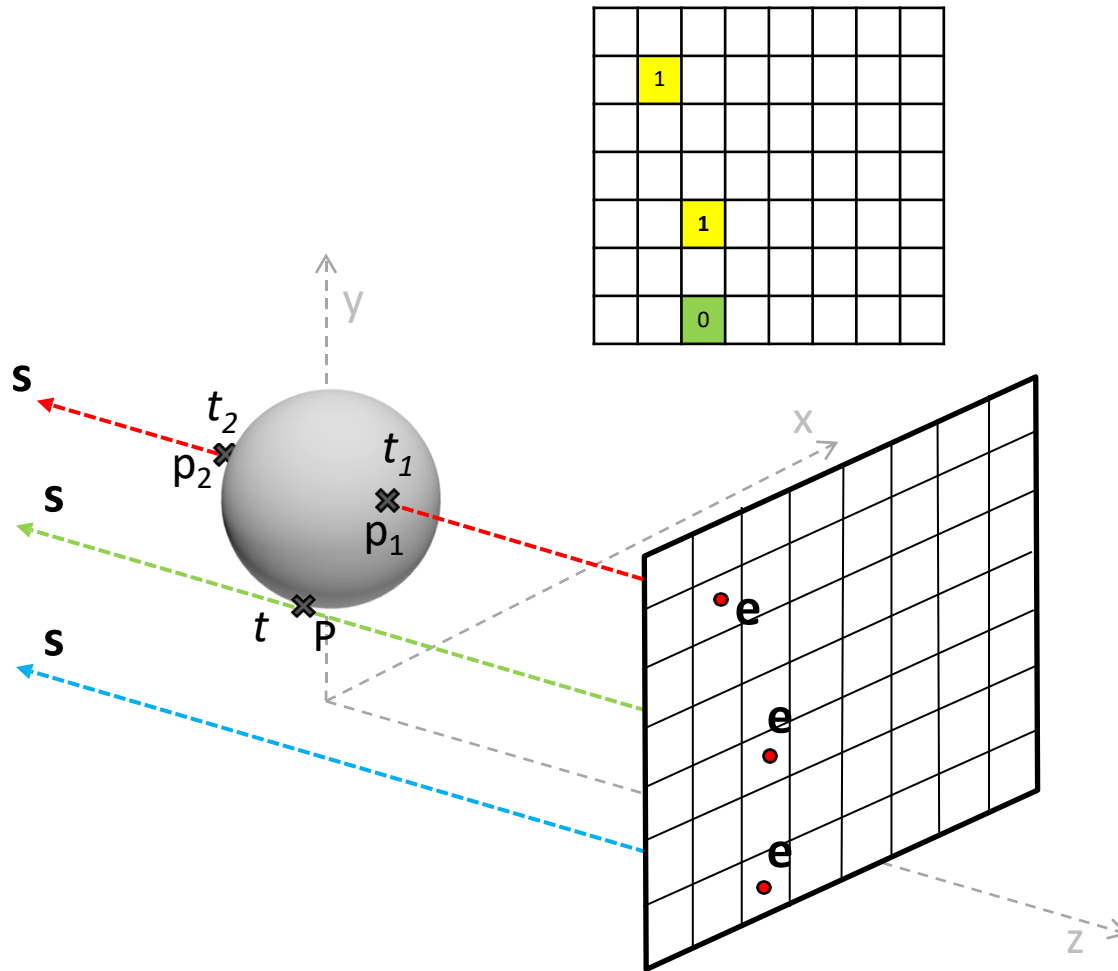
# Ray - Sphere Intersection (5/8)

$$t = \frac{-B \pm \sqrt{B^2 - 4AC}}{2A}$$

- $B^2 - 4AC$ , is called
  - **negative**: its square root is imaginary, sphere **do not intersect**
  - **positive**: there are two solutions
    - one solution where the ray enters the sphere
    - one where it leaves the sphere
  - **zero**: the ray grazes the sphere at a single **point**.



# Ray - Sphere Intersection (6/8)



|  |   |   |  |  |  |  |
|--|---|---|--|--|--|--|
|  |   |   |  |  |  |  |
|  | 1 |   |  |  |  |  |
|  |   |   |  |  |  |  |
|  |   | 1 |  |  |  |  |
|  |   |   |  |  |  |  |
|  |   | 0 |  |  |  |  |

$$\begin{aligned} \mathbf{p} &= \mathbf{e} + t(\mathbf{s} - \mathbf{e}) \\ &= \mathbf{e} + t\mathbf{d} \end{aligned}$$

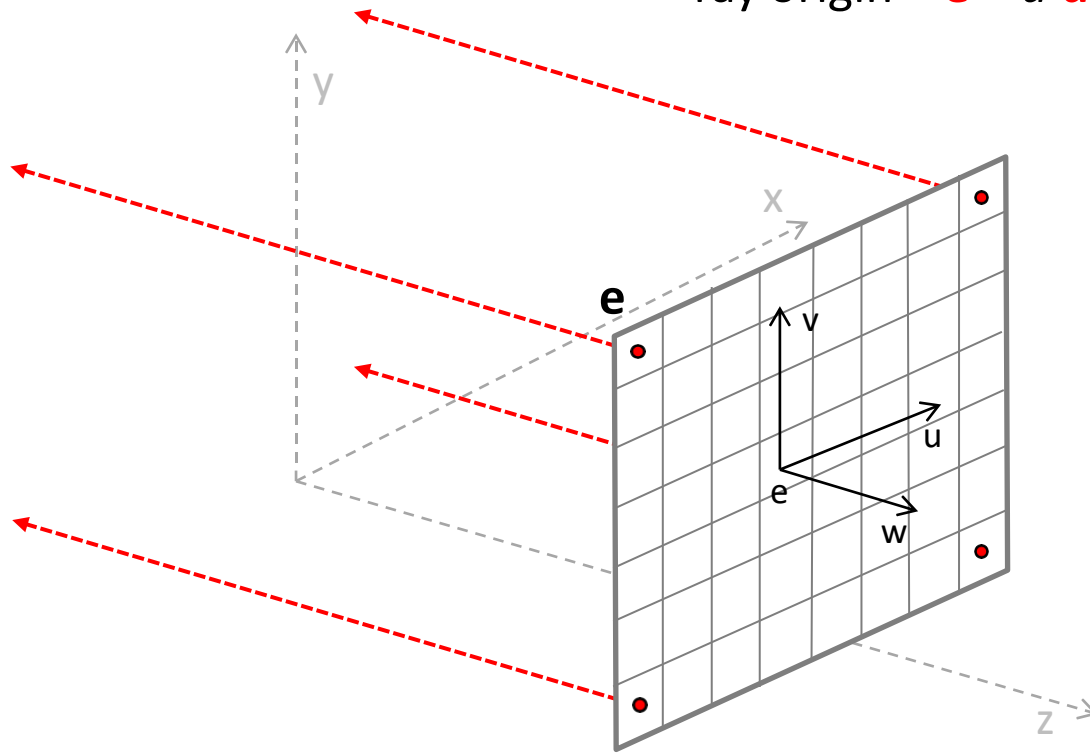
$$t = \frac{-B \pm \sqrt{B^2 - 4AC}}{2A}$$



# Ray - Sphere Intersection (7/8)

Orthographic:

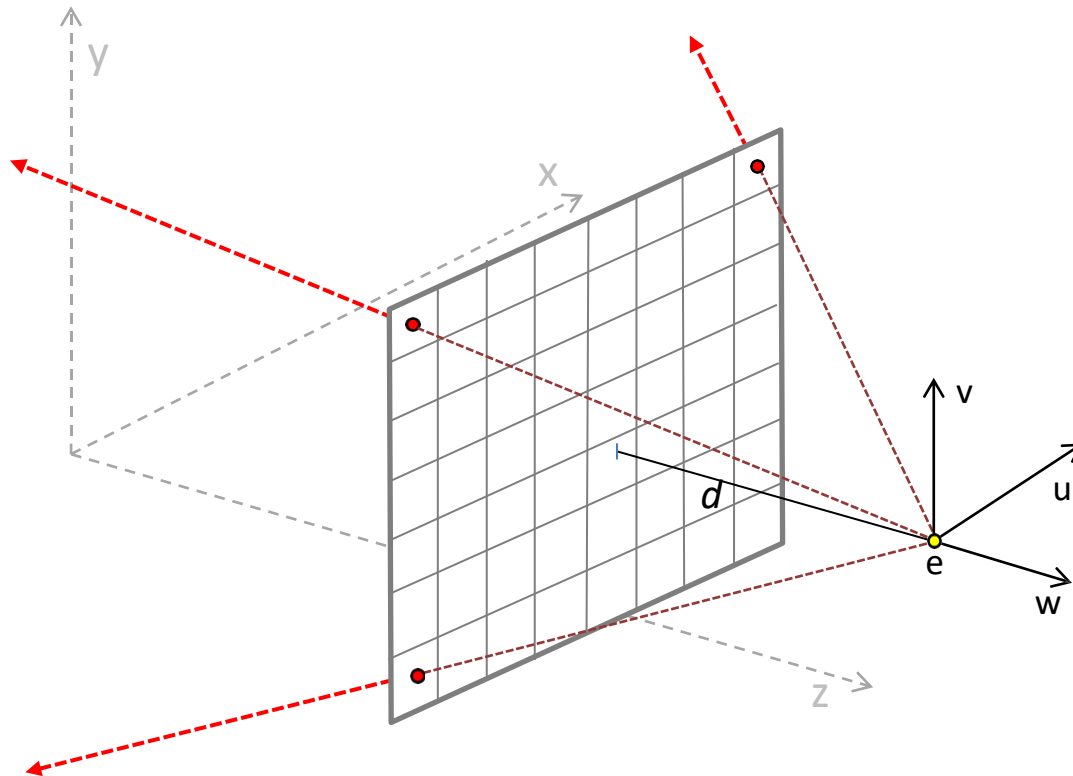
- ray direction =  $-\mathbf{w}$
- ray origin =  $\mathbf{e} + u \mathbf{u} + v \mathbf{v}$



# Ray - Sphere Intersection (8/8)

Perspective:

- ray direction =  $-d\mathbf{w} + u\mathbf{u} + v\mathbf{v}$
  - ray origin =  $\mathbf{e}$
- Q: why?*

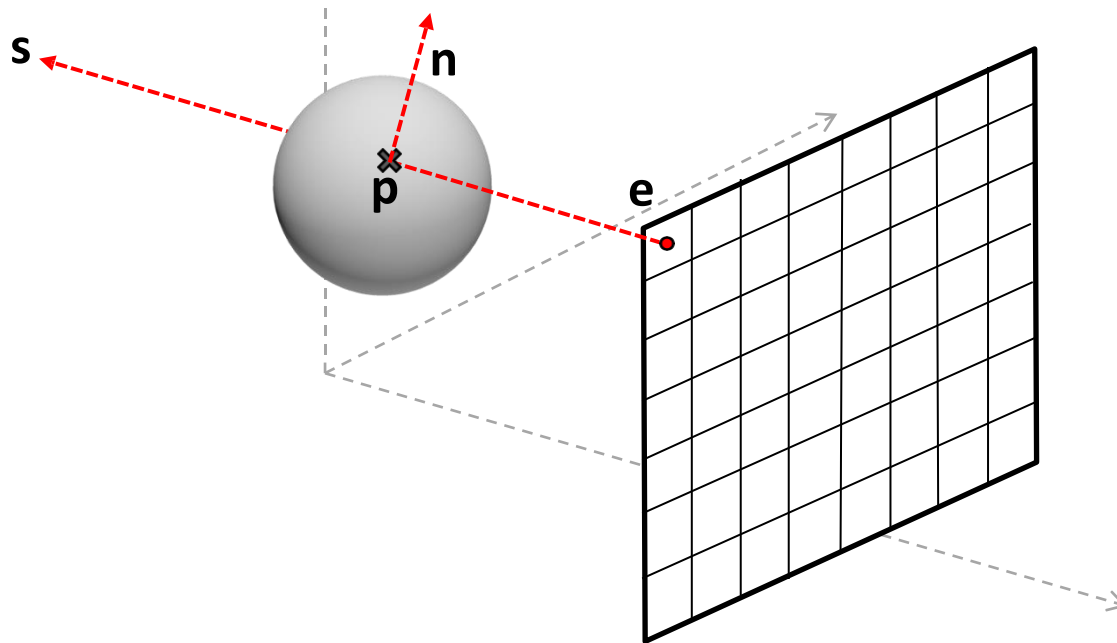


# Shading (1/3)

Normal vector at point **p**:

- Gradient,  $\mathbf{n} = 2 (\mathbf{p} - \mathbf{c})$ .
- unit normal is  $(\mathbf{p} - \mathbf{c})/R$ .

*[See section 2.5.4]*

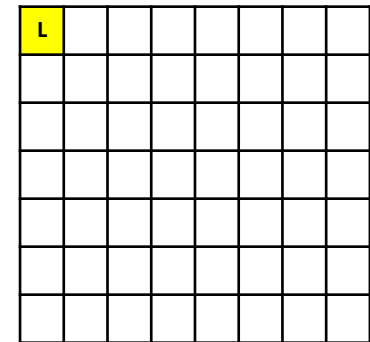
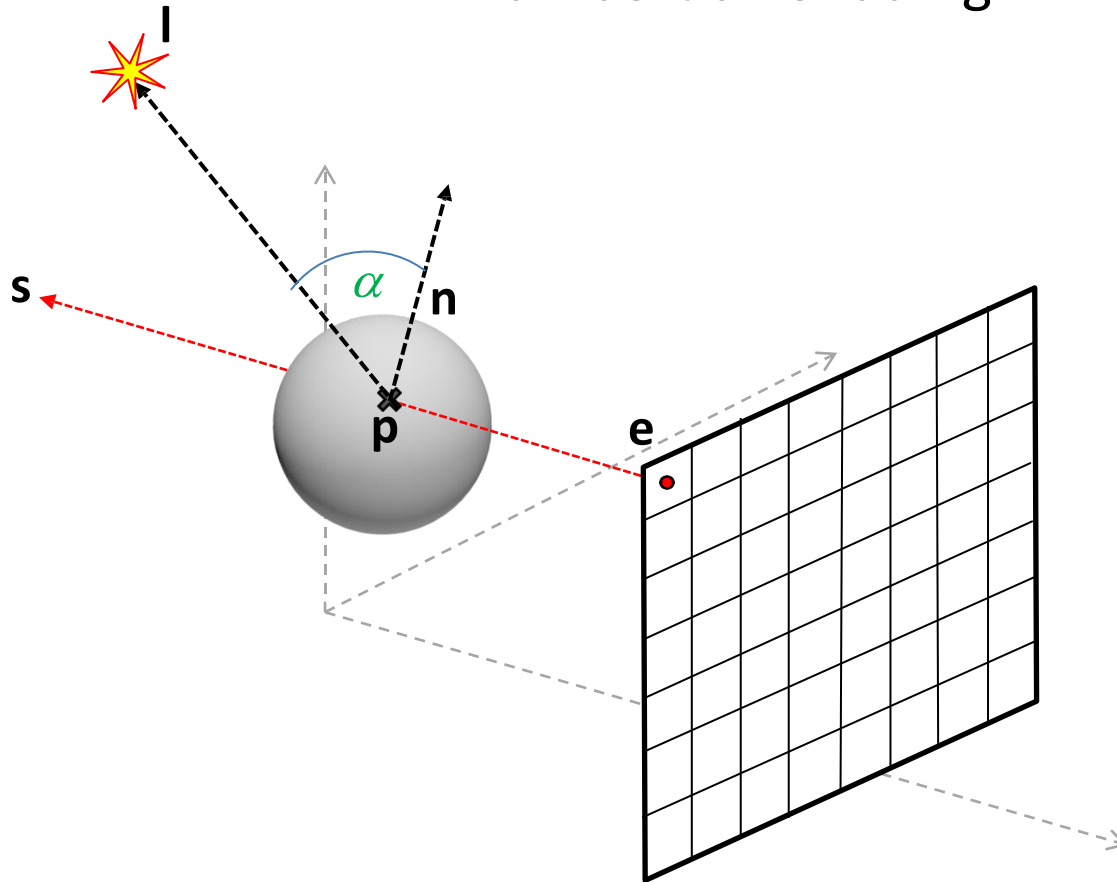


# Shading (2/3)

Lambertian Shading:  $L = k_d P \max(0, \mathbf{n} \cdot \mathbf{l})$

where,

- $L$  = pixel color
- $k_d$  = surface color
- $P$  = intensity of the light source.



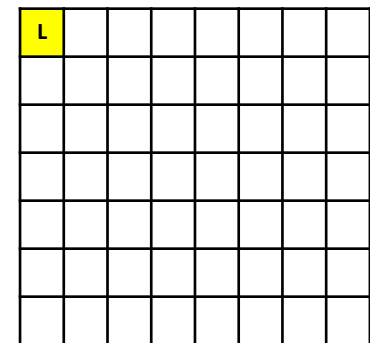
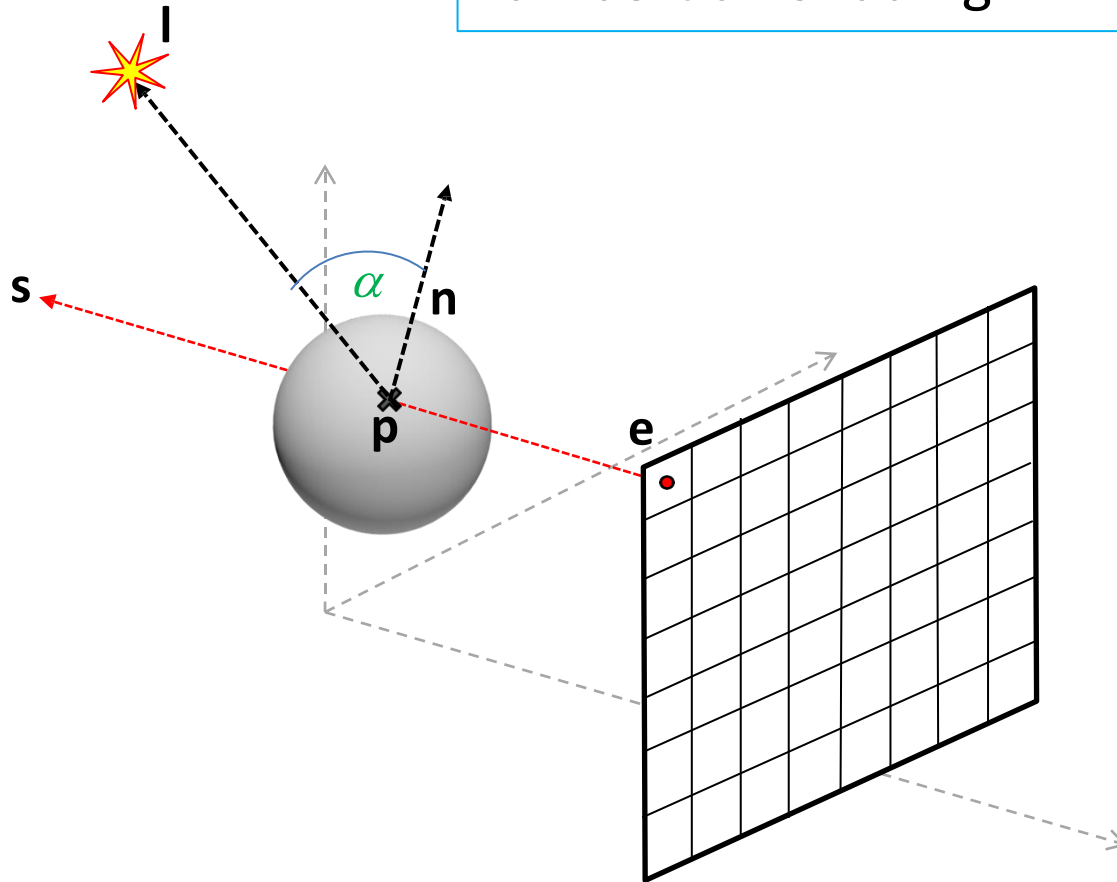
# Shading (3/3)

Q: Are we considering angle in this formula? If yes – how?

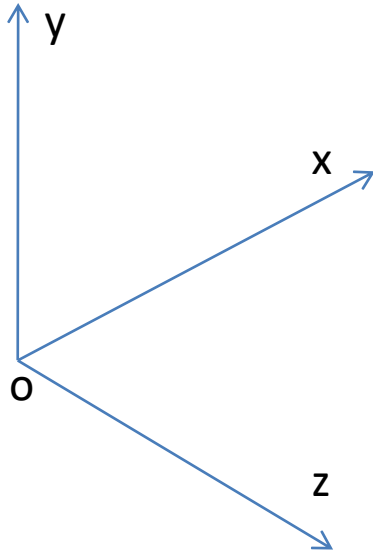
Lambertian Shading:  $L = k_d P \max(0, \mathbf{n} \cdot \mathbf{l})$

where,

- $L$  = pixel color
- $k_d$  = surface color
- $P$  = intensity of the light source.



# Practice Problem (1/3)



**Camera frame (*orthographic*):**

- $\mathbf{e} = [4, 4, 6]$ ;  $\mathbf{u} = [1, 0, 0]$ ;  $\mathbf{v} = [0, 1, 0]$ ;  $\mathbf{w} = [0, 0, 1]$ 
  - Plot the camera frame on the given axis.

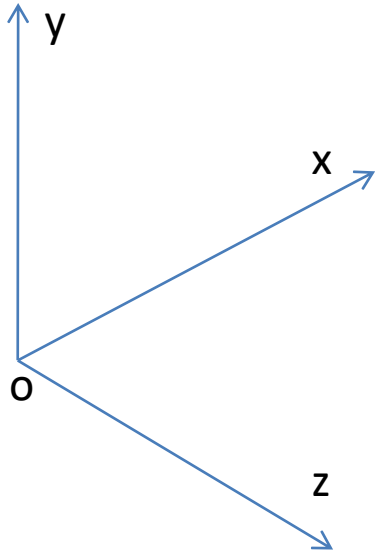
**Viewing Ray:**

- $\text{ray}_1.\text{origin} = \mathbf{e} + 2\mathbf{u} + 2\mathbf{v}$ ;  $\text{ray}_1.\text{end} = [6, 6, 0]$
- $\text{ray}_2.\text{origin} = \mathbf{e} - 1\mathbf{u} + 1\mathbf{v}$ ;  $\text{ray}_2.\text{end} = [4, 4, 0]$ 
  - Plot the origins for  $\text{ray}_1$  and  $\text{ray}_2$ .

**Sphere:**  $f(x, y, z) = x^2 + y^2 + z^2 - (4)^2 = 0$

1. What are the intersecting points for  $\text{ray}_1$  and  $\text{ray}_2$ ?
2. Plot the intersecting points.

# Practice Problem (2/3)



**Camera frame (*orthographic*):**

- $\mathbf{e} = [4, 4, 8]$ ;  $\mathbf{u} = [1, 0, 0]$ ;  $\mathbf{v} = [0, 1, 0]$ ;  $\mathbf{w} = [0, 1, 0]$

**Image Plane:**

- left:  $u = -5$ ; right:  $u = 5$ ; top:  $v = 4$ ; bottom:  $v = -4$

1. Plot the image plane on the given axis.
2. For a  $10 \times 10$  image matrix  $M$ , what is the position on the image plane for the ray origin at  $M(4,3)$ ?
3. Will it intersect  $f(x, y, z) = x^2 + y^2 + z^2 - 5^2 = 0$ ?

# Practice Problem (3/3)

Consider the following parameters for an orthographic ray-tracing:

- *Camera frame:*

$$E = [-2, 7, 17]^T, U = [1, 0, 0]^T, V = [0, 1, 0]^T, W = [0, 0, 1]^T$$

- *Image plane:*

$$l = -15, r = 15, t = 10, b = -10$$

- *Raster image resolution:*  $13 \times 11$
- *Sphere:*  $(x+3)^2 + (y-5)^2 + (z-3)^2 = 64$

Determine the ray-sphere intersection point(s) for a ray (with *length* = 25) at the *center* of the raster image. Drawing figures is NOT mandatory.



# Practice Problem (3/3)

## Solution steps:

- Find  $u$  and  $v$   
$$u = l + (r - l)(i + 0.5) / nx$$
$$v = b + (t - b)(j + 0.5) / ny$$
- Determine the ray origin,  $\mathbf{e} = \mathbf{E} + u\mathbf{U} + v\mathbf{V}$
- Find ray end point,  $\mathbf{s} = \mathbf{e} + w(-\mathbf{W})$
- Determine,  $\mathbf{d} = \mathbf{s} - \mathbf{e}$
- Determine,  $D = B^2 - 4AC$   
$$A = \mathbf{d} \cdot \mathbf{d}$$
$$B = 2\mathbf{d} \cdot (\mathbf{e} - \mathbf{c})$$
$$C = (\mathbf{e} - \mathbf{c}) \cdot (\mathbf{e} - \mathbf{c}) - R^2$$
- Determine the intersection parameter,  $t$   
$$t_1 = (-B + \sqrt{D}) / (2A)$$
$$t_2 = (-B - \sqrt{D}) / (2A)$$
- Determine the intersection point,  
$$\mathbf{P}_1 = \mathbf{e} + t_1(\mathbf{s} - \mathbf{e})$$
$$\mathbf{P}_2 = \mathbf{e} + t_2(\mathbf{s} - \mathbf{e})$$

# Further Reading

- Fundamentals of Computer Graphics, 4th Edition -  
Chapter 4